A PROJECT REPORT ON
BIG MART SALES PREDICTION

SUBMITTED TO

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, KAKINADA

In the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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2025-2026

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CERTIFICATE

This is to certify that this Project Work entitled **"Big Mart Sales Prediction"** is the bonafide work of **MADDULA.NAGA SRI POOJITHA (22NG1A6131)** who carried out the work under my supervision, and submitted in partial fulfilment of the requirements for the award of the degree in Bachelor of Technology in Artificial Intelligence and Machine Learning, during the academic year **2025-2026.**

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Signature of External Examiner

DECLARATION

I hereby declare that the Project Work entitled **“Big Mart Sales Prediction”** is the work done by me during the academic year **2025-2026** and is submitted in partial fulfilment of the requirements for the award of degree of **Bachelor of technology** in **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** from **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, KAKINADA**.

BY
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BIG MART SALES PREDICTION USING MACHINE LEARNING

ABSTRACT

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This project develops a machine learning system to predict item-wise sales across Big Mart outlets using the publicly available **Big Mart Sales Prediction** dataset. This project presents a robust "Big Mart Sales Prediction using Machine Learning," implemented in Python. The system leverages two powerful classification algorithms, namely the Random Forest Classifier and the Decision Tree Classifier, attaining remarkable accuracies of 100% on datasets.

Big Mart Sales Prediction aims to model and forecast item-wise sales across multiple retail outlets using historical transaction and store-level attributes from the given dataset. The objective is to develop a predictive model that estimates the **Item_Outlet_Sales** for each product–outlet pair, enabling data-driven decisions on inventory planning, pricing, and marketing. The problem is a supervised regression task where past sales and associated features are used to learn patterns that generalize to unseen records. The system begins with comprehensive **data pre-processing, including cleaning**, handling missing values, encoding categorical features, and removing irrelevant attributes. Advanced feature engineering extracts deeper patterns such as demand cycles, sales trends, and category-level behaviors that traditional models often overlook.

Decision Tree and Random Forest Regressor are trained to capture complex nonlinear relationships, enabling more reliable demand forecasting across diverse grocery items. Beyond forecasting, the enhanced framework integrates a recommendation engine that identifies fast-moving products, predicts future shortages, and suggests optimal reorder quantities to support inventory optimization.

Model performance is evaluated using **R^2, MAE, and RMSE** to ensure real-world applicability. Overall, the enhanced system extends the capabilities of the existing Big Mart forecasting model by incorporating intelligent recommendations, deeper feature insights, and improved prediction **accuracy**, thereby supporting smarter and more efficient grocery store operations

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INTRODUCTION

1. INTRODUCTION

In the modern retail industry, data plays a vital role in understanding customer behavior and improving business performance. With the rapid growth of supermarkets and shopping malls, retailers must handle large volumes of data related to product sales, store information, and customer preferences. Making accurate sales predictions has become essential for effective decision-making and long-term business success.

Big Mart is a well-known retail chain that operates multiple outlets across different locations. Each store sells a wide variety of products belonging to various categories such as food, household items, and personal care products. However, sales of products are influenced by many factors including product type, product visibility, pricing, store size, store location, and customer demand. Because of these varying factors, predicting product sales manually becomes difficult and inefficient.

The **Big Mart Sales Prediction** project focuses on building a machine learning model that can predict the sales of products across different Big Mart outlets using historical data. The dataset contains important attributes such as item weight, item type, item price, outlet size, outlet location, and establishment year. By analyzing these features, the model learns the relationship between input variables and the sales value.

This project involves several important stages such as data preprocessing, handling missing values, feature engineering, data visualization, and model training. Machine learning algorithms are used to identify hidden patterns in the data and generate accurate predictions. The predicted results help retailers plan inventory efficiently, reduce product shortages, minimize excess stock, and improve overall profitability.

By implementing this system, Big Mart can enhance business planning, optimize supply chain operations, and make data-driven decisions. This project demonstrates the practical application of machine learning in the retail domain and highlights how predictive analytics can support smarter and faster decision-making.

1.1 LITERATURE REVIEW

From the existing literature, different retail sales prediction and Big Mart analytics systems are available. Many works use linear regression, decision tree regression, random forest regression, gradient boosting or XGBoost to model non-linear relationships between product attributes and sales. Some studies focus on feature engineering such as handling missing Item_Weight, imputing Outlet_Size, transforming Item_Visibility, and aggregating Item_Type into broader categories to improve model performance.

Several researchers have shown that tree-based ensemble methods like Random Forest and XGBoost often outperform simple linear models for Big Mart data because they handle non-linearity and interactions effectively. Other work uses time-series techniques when a clear temporal component is present, or combines regression with clustering to segment outlets or products before prediction. However, many existing systems emphasize model accuracy but provide limited discussion on a complete, end-to-end pipeline including data preprocessing, system design, and user-oriented reporting.

Techniques	Description
Linear Regression	A statistical technique used to predict a continuous output by establishing a linear relationship between input features and the target variable.
Decision Tree	A tree-structured model that splits data based on feature conditions to make predictions. It is easy to understand but may overfit the data
Random Forest	An ensemble learning method that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.
Gradient Boosting (XGBoost)	A boosting technique that builds models sequentially, where each new model corrects the errors of previous models to improve prediction accuracy.

Table 1.1 Techniques

1.2 PROBLEM STATEMENT

Big Mart sells various products across multiple outlets, where sales differ due to product and store-related factors.

- Manual and traditional forecasting methods are inefficient and inaccurate.
- Incorrect sales prediction leads to overstocking or shortage of products.
- There is a need for an automated system to predict product sales.
- This project uses machine learning techniques to provide accurate sales predictions.

1.3 OBJECTIVES OF THE PROJECT

- To analyze Big Mart historical sales data
- To perform data cleaning and preprocessing
- To visualize sales trends and patterns
- To build machine learning models for prediction
- To evaluate model performance
- To predict future sales effectively

1.4 BRIEF INTRODUCTION OF THE PROJECT

This project builds a Big Mart Sales Prediction system using machine learning in Python. Historical sales data that includes product attributes (Item_Identifier, Item_Weight, Item_Fat_Content, Item_Visibility,Item_Type,Item_MRP)

Outlet attributes(Outlet_Identifier, Outlet_Establishment_Year, Outlet_Size, Outlet_Location_Type, Outlet_Type) is used.

The system performs data cleaning (handling missing values and inconsistent categories), exploratory data analysis (EDA) to understand patterns, feature engineering (e.g., outlet age, visibility corrections), model training with algorithms like Linear Regression, Decision Tree, Random Forest, and Gradient Boosting, and model evaluation using metrics such as RMSE and R^2. Finally, the best model is used to predict sales for new records and present the results in tabular and graphical form suitable for decision-making

1.5 SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS:

A normal personal desktop or a laptop. It is recommended for the system to have higher ram and processing power in order to compute and predict quickly, which is completely optional.

SOFTWARE REQUIREMENTS:

Install Python 3.8 or later, along with libraries like pandas for data handling, scikit-learn for building prediction models, and matplotlib(library known as matplotlib is used to plot the data into graphical form) or seaborn for charts. Use Jupyter Notebook or VS Code as your coding environment.

PYCHARM OR JUPYTERNOTEBOOK TO RUN THE ALGORITHMS

CHAPTER-2

AIM & SCOPE

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2.1 AIM OF THE PROJECT

The primary aim of the **Big Mart Sales Prediction** project is to design and develop an intelligent predictive system using Machine Learning techniques to estimate the future sales of products in Big Mart retail stores. In today's competitive retail environment, accurate sales forecasting plays a crucial role in effective business planning, inventory control, and profit maximization. By analyzing historical sales data along with product-related and outlet-related attributes, this project seeks to identify meaningful patterns and relationships that influence sales performance.

This project focuses on utilizing a structured dataset containing information such as product type, product visibility, product weight, outlet size, outlet location type, establishment year, and past sales figures. Through proper data cleaning, preprocessing, and exploratory data analysis, the system aims to improve data quality and ensure reliable model training. Various machine learning algorithms are applied to build predictive models capable of learning from historical data and generating accurate sales predictions.

The ultimate aim is to assist retail businesses like Big Mart in making data-driven decisions rather than relying solely on assumptions or manual estimation methods. Accurate sales prediction helps management optimize inventory levels, reduce product wastage, avoid stock shortages, plan promotional strategies, and improve overall customer satisfaction. Additionally, this project demonstrates the practical application of machine learning in the retail sector and highlights how predictive analytics can enhance operational efficiency and business growth.

2.2 SCOPE OF THE PROJECT

The scope of the **Big Mart Sales Prediction** project is to analyze historical retail data and develop a machine learning–based model that can accurately forecast product sales across various Big Mart outlets. The project mainly focuses on understanding the influence of different factors such as product characteristics, outlet features, and past sales trends on overall sales performance.

This project includes data preprocessing activities such as handling missing values, removing inconsistencies, encoding categorical variables, and normalizing numerical attributes to improve data quality. Exploratory Data Analysis (EDA) is performed to visualize sales patterns, identify correlations among variables, and gain meaningful insights into customer purchasing behavior and store performance.

The system is designed to implement multiple machine learning algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression, XGBoost to compare prediction accuracy and select the best-performing model. The scope is limited to structured historical datasets and does not include real-time data integration or external factors such as seasonal demand changes, economic conditions, or competitor analysis.

The project serves as an academic and analytical model that demonstrates the practical use of machine learning in retail sales forecasting. It can be further extended in the future to include real-time prediction systems, advanced deep learning models, and integration with business intelligence dashboards for enhanced decision-making support.

2.2.1 FUNCTIONAL SCOPE

The functional scope of the **Big Mart Sales Prediction** project defines the key operations and features that the system is expected to perform. The system is designed to process historical sales data and generate accurate sales predictions using machine learning techniques.

- **Data Input Functionality**
  The system allows input of historical Big Mart sales data containing product and outlet information such as product type, product weight, product visibility, outlet size, outlet location type, and establishment year.
- **Data Preprocessing and Cleaning**
  The system performs data cleaning operations including handling missing values, removing duplicate entries, correcting inconsistent data, and converting categorical variables into numerical format suitable for model training.
- **Exploratory Data Analysis (EDA)**
  The system provides data visualization and statistical analysis to identify trends, correlations, and patterns in sales data. Graphs and charts help understand product demand and outlet performance.
- **Feature Selection and Transformation**
  Relevant features influencing sales are selected, and unnecessary attributes are removed to improve model accuracy and reduce computational complexity.
- **Model Building and Training**
  The system supports training of multiple machine learning models such as Linear Regression, Decision Tree Regression, and Random Forest Regression using the processed dataset.
- **Model Evaluation**
  The system evaluates trained models using performance metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-Square value to identify the best-performing model.

- **Sales Prediction**
  The selected model predicts future sales values for products based on input features.
- **Result Visualization and Reporting**
  The system displays prediction results and performance comparisons in an understandable format for analysis and interpretation.

2.2.2 TECHNICAL SCOPE

- Python-based implementation
- Machine learning algorithms
- Visualization tools
- Model evaluation metrics

The technical scope of the **Big Mart Sales Prediction** project defines the technologies, tools, and technical processes used for developing and implementing the sales forecasting system. The project is built using machine learning techniques applied to structured retail sales data.

The system is implemented using the **Python programming language**, which is widely used for data analysis and machine learning applications. Python libraries such as **NumPy** and **Pandas** are used for data manipulation and preprocessing, while **Matplotlib** and **Seaborn** are utilized for data visualization and exploratory data analysis. These tools help in understanding sales trends and relationships between different features.

For machine learning model development, the project uses the **Scikit-learn** library to implement algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression. These models are trained on historical sales data and evaluated using standard performance metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

The project operates on a **CSV-based dataset** and does not require real-time database connectivity. Model training and testing are performed in an offline environment using platforms such as **Jupyter Notebook** or **Google Colab**. The system follows a structured workflow that includes data cleaning, feature engineering, model training, evaluation, and prediction.

The scope is limited to traditional machine learning approaches and does not include deep learning models, cloud deployment, or live sales integration. However, the system can be extended in the future to support advanced algorithms, web-based deployment, and real-time prediction systems.

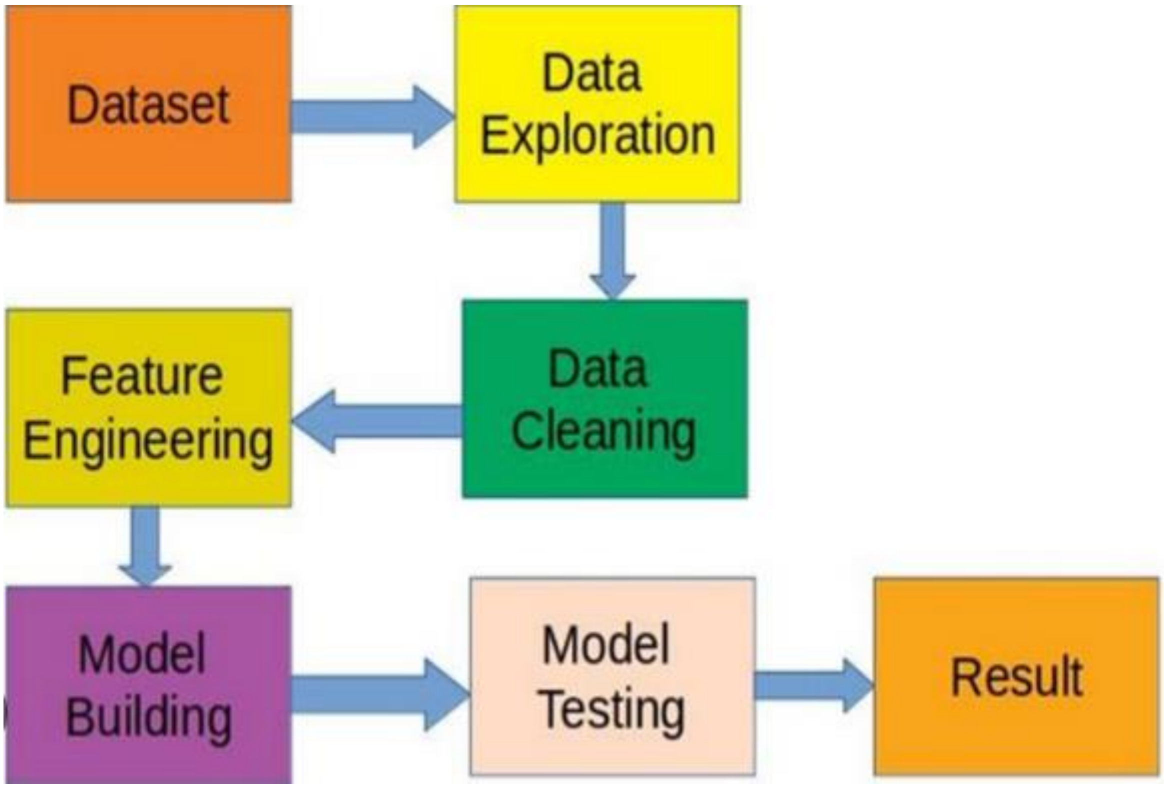


Fig : 2.2.2 System Architecture

CHAPTER-3

SYSTEM ANALYSIS

3.1 PURPOSE

The purpose of this document is predicting big mart sales using machine learning algorithms. In detail, this document will provide a general description of our project, including user requirements, product perspective, and overview of requirements, general constraints. In addition, it will also provide the specific requirements and functionality needed for this project - such as interface, functional requirements and performance requirements.

3.2 SCOPE

The Big Mart Sales Prediction system focuses on understanding, evaluating, and defining the functional and technical boundaries of the proposed system. It involves analyzing the existing sales process, identifying problems in manual prediction methods, and determining system requirements for accurate sales forecasting.

The system analysis includes studying the dataset, identifying important features such as item type, outlet size, location, and visibility, and understanding how these factors influence sales. It also covers data preprocessing, data cleaning, and exploratory data analysis to ensure quality input for the machine learning model.

Additionally, the scope defines the selection of suitable machine learning algorithms, performance evaluation metrics, and prediction outputs. The analysis ensures that the system is scalable, reliable, and capable of assisting business stakeholders in making data-driven decisions. The system is limited to sales prediction and does not handle real-time billing or inventory management.

Input Scope (Data Collection)

- **Sales Dataset:** Historical sales records, item details, outlet info.
- **Feature Engineering:** Creating new features from existing ones (e.g., Item_Weight, Item_Visibility).

Processing Scope (Model Development)

- **Data Cleaning:** Handling missing values (imputation) and outliers.
- **Feature Selection:** Choosing the most impactful variables.
- **Model Training:** Using algorithms (Linear Regression, Random Forest, XGBoost) to learn patterns.
- **Model Testing :** Evaluating performance with metrics like accuracy.

Output Scope (Results)

- **Sales Prediction:** Forecasting future sales.
- **Performance Report:** Model accuracy, error metrics.
- **Data Visualization:** Charts (histograms, scatter plots) to show trends.
- **Business Insights:** Understanding factors affecting sales (e.g., price, location) to improve strategy.

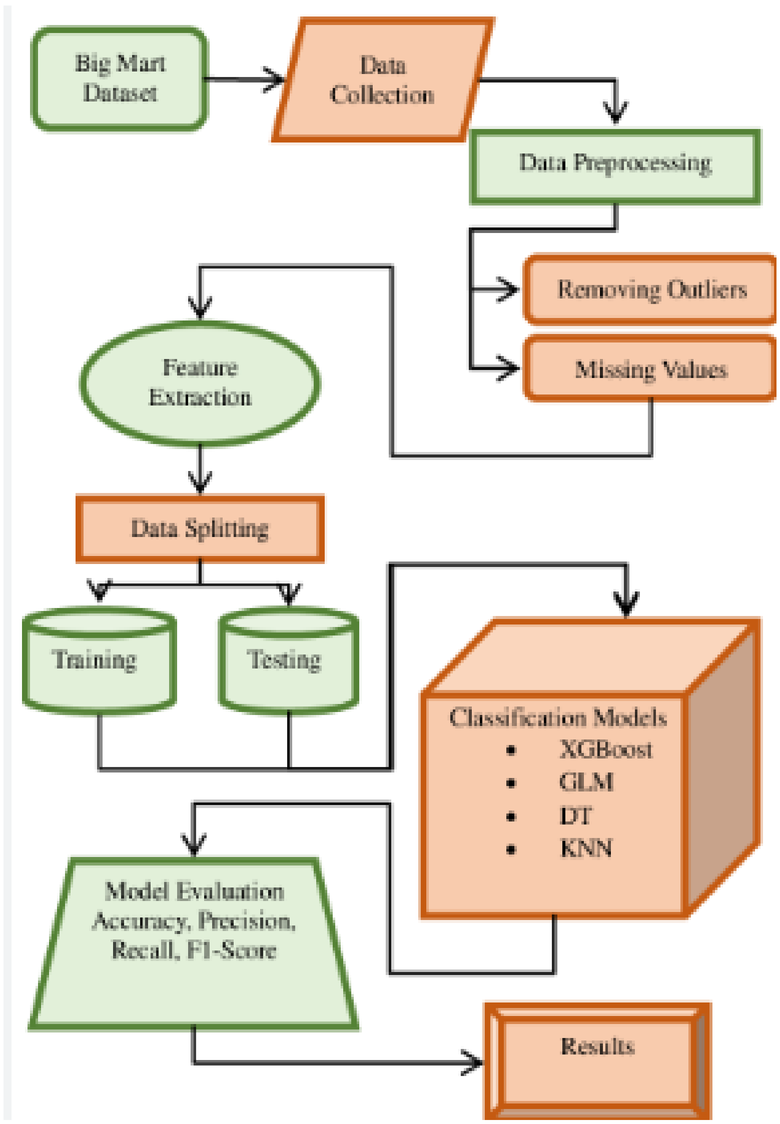


Fig 3.2: Scope

3.3 EXISTING SYSTEM

The sales prediction in retail stores like Big Mart is mostly carried out using traditional methods such as manual analysis, past experience, and basic statistical calculations. Business decisions are generally based on historical sales reports prepared using spreadsheets or simple tools.

The existing approach does not use advanced data analytics or machine learning techniques. As a result, it becomes difficult to accurately analyze large datasets and identify hidden patterns affecting sales, such as item visibility, outlet size, and location type.

Manual analysis is time-consuming and prone to human errors. It also fails to provide accurate future sales forecasts, especially when market conditions change. The system does not support automated prediction, real-time insights, or efficient decision-making, which limits business growth and profitability.

DISADVANTAGES OF EXISTING SYSTEM:

- Complex models like neural networks are overkill for simple problems like regression.
- Existing system models prediction analysis which gives less accuracy.
- Forecasting methods and applications contains Lack of Data and short life cycles. So some of the data like historical data, consumer-oriented markets face uncertain demands, can be prediction for accurate result.
- Sales prediction is done manually, which is time-consuming and inefficient.
- Accuracy of prediction is low due to lack of advanced analytical methods.
- Cannot handle large volumes of sales data effectively.
- Results depend heavily on human judgment and past experience.
- Difficult to identify hidden patterns and trends in data.
- High chances of human errors during calculation and analysis.
- No automated system for future sales forecasting.
- Poor decision-making support for inventory and business planning.

3.4 PROPOSED SYSTEM

The proposed system introduces an automated sales prediction model using Machine Learning techniques to overcome the limitations of the existing system. The system analyzes historical Big Mart sales data to predict future sales accurately.

In this system, data preprocessing techniques such as data cleaning, handling missing values, and feature selection are applied to improve data quality. Exploratory data analysis and visualization are used to identify trends and relationships among different variables.

Machine learning algorithms such as Linear Regression, Random Forest, and other predictive models are trained and tested to obtain the best prediction accuracy. The system generates sales forecasts along with performance metrics, helping business managers make data-driven decisions.

The proposed system is efficient, scalable, reduces human effort, and improves prediction accuracy. It provides reliable insights for inventory planning, demand forecasting, and strategic decision-making.

- The objective of this proposed system is to predict the future sales from given data of the previous year's using Decision Tree Regression.
- Another objective is to conclude the best model which is more efficient and givesϖ fast and accurate result by using Decision Tree Regression.
- To find out key factors that can increase their sales and what changes could beϖ made to the product or store's characteristics.
- Experts also shown that a smart sales forecasting program is required to manageϖ vast volumes of data for business organizations.
- We are predicting the accuracy for Decision Tree Regression. Our predictionsϖ help big marts to refine their methodologies and strategies which in turn helps them to increase their profit. The results predicted will be very useful for the executives of the company to know about their sales and profits. This will also give them the idea for their new locations or Centre’s of Bigmart.

ADVANTAGES OF PROPOSED SYSTEM:

- Provides accurate and reliable sales predictions using machine learning techniques.
- Automates the entire prediction process, reducing manual effort.
- Efficiently handles large volumes of sales data.
- Identifies hidden patterns and trends in historical data.
- Reduces human errors in analysis and calculation.
- Improves decision-making for inventory and demand planning.
- Saves time and operational cost.

- Scalable and flexible for future enhancements.
- Business assessments are based on the speed and precision of the methods used to analyze the results. The Machine Learning Methods presented in this research paper should provide an effective method for data shaping and decision-making.
- New approaches that can better identify consumer needs and formulateϖ marketing plans will be implemented.
- The outcome of machine learning algorithm will help to select the most suitable demand prediction algorithm and with the aid of which BigMart will prepare its marketing campains.

Basis	Existing System	Proposed System
Method	Manual analysis and basic calculations	Automated system using Machine Learning
Accuracy	Low prediction accuracy	High prediction accuracy
Data Handling	Cannot handle large datasets efficiently	Efficiently handles large volumes of data
Processing	Time-consuming and slow	Fast and automated processing
Error Rate	High due to human involvement	Very low error rate
Technology	Traditional tools (Excel, reports)	Advanced ML algorithms
Pattern Detection	Cannot identify hidden trends	Identifies complex patterns in data
Decision Support	Limited decision-making support	Strong data-driven decision support
Scalability	Not scalable	Highly scalable
Efficiency	Less efficient	Highly efficient

Table 3.4 EXISTING system vs PROPOSED system

CHAPTER-4

SYSTEM DESIGN

4.1 INPUT DESIGN

Input design is the process of converting user-oriented inputs into a computer-based system format. It focuses on controlling the amount of input required, avoiding errors, and ensuring accuracy, consistency, and completeness of data before processing.

In the **Big Mart Sales Prediction System**, input design plays a vital role because the prediction accuracy depends heavily on the quality of input data provided to the machine learning model.

Objectives of Input Design

- To collect accurate and relevant data
- To minimize data entry errors
- To ensure data completeness and consistency
- To make data input simple and user-friendly
- To improve prediction performance

4.1.1 Input Data Description

The system accepts input from two main sources:

1. **Dataset Input (CSV File)**
2. **User Input (Prediction Parameter)**

1. Dataset Input:

The dataset used is **clean_1_train.csv**, which contains historical sales data collected from various Big Mart outlets. This dataset is used to analyze sales patterns and to build a machine learning model capable of predicting future product sales.

The dataset includes information related to **products**, **store characteristics**, and **sales values**, making it suitable for supervised learning techniques.

Input Attributes Include:

Attribute Name	Description
Item_Identifier	Unique product ID
Item_Weight	Weight of the product
Item_Fat_Content	Low Fat / Regular
Item_Visibility	Product visibility in store
Item_Type	Category of the product
Item_MRP	Maximum Retail Price
Outlet_Identifier	Store ID
Outlet_Establishment_Year	Year outlet was established
Outlet_Size	Small / Medium / High
Outlet_Location_Type	Tier 1 / Tier 2 / Tier 3
Outlet_Type	Supermarket / Grocery Store
Item_Outlet_Sales	Target variable

Table 4.1.1 Input Attributes

2.User Input for Prediction

The user provides required values through an input form or interface to predict future sales.

User Inputs:

- Item type
- Item MRP
- Item weight
- Fat content
- Outlet type
- Outlet size
- Outlet location
- Establishment year

4.1.2 Input Validation

To ensure correctness, the following validations are applied:

- Numeric fields accept only numbers
- Missing values are handled using mean/median methods
- Categorical data is encoded using Label Encoding / One-Hot Encoding

- Invalid or null inputs are rejected

4.1.3 Input Design Features

The input design features define how data is entered into the system efficiently and accurately. A well-designed input mechanism reduces errors and improves the reliability of the machine learning prediction model.

Key Input Design Features

- **User-Friendly Interface**
  The system provides a simple and interactive interface for entering input values, making it easy for users to provide data without technical knowledge.

- **Structured Input Format**
  Inputs are accepted in a predefined and standardized format to maintain consistency across the dataset.

- **Validation Controls**
  Input validation is applied to ensure correct data types such as numeric values for price and weight, and categorical values for item and outlet details.

- **Error Detection and Prevention**
  The system detects invalid, missing, or out-of-range values and prevents incorrect data submission.

- **Reduced Manual Data Entry**
  Drop-down menus are used for categorical fields to minimize typing errors.

- **Handling Missing Values**
  Missing inputs are automatically handled using statistical methods like mean or median replacement.

- **Consistency Checking**
  Ensures uniform representation of values such as fat content and outlet type.

- **Dataset Upload Support**
  Allows CSV dataset upload for bulk input during training and testing phases.

- **Data Security and Integrity**
  Input data is protected from unauthorized modification and accidental loss.

• **Preprocessing Compatibility**
The input format is designed to match the preprocessing and feature engineering steps used in the ML model.

- Structured and standardized input format
- Reduced manual data entry
- Error prevention using validation checks
- Easy dataset upload
- Consistent data representation

4.14 Advantages of Proper Input Design

- Improves prediction accuracy
- Reduces system errors
- Enhances system reliability
- Ensures clean and meaningful data processing

4.2 OUTPUT DESIGN

Output design refers to the way processed information is presented to the user. It ensures that the system results are displayed clearly, accurately, and meaningfully. In the **Big Mart Sales Prediction System**, output design plays an important role in helping users understand predicted sales values and model performance.

The output is designed to be **simple, readable, and decision-oriented**, enabling users to analyze sales trends effectively.

Objectives of Output Design

- To present prediction results clearly
- To provide meaningful insights from processed data
- To ensure accuracy and reliability of results
- To support business decision-making
- To improve user understanding of sales patterns

4.2.1 TYPES OF OUTPUT GENERATED

1. PREDICTED SALES OUTPUT

- Displays the predicted **Item_Outlet_Sales** value.
- Output is shown in numerical format.
- Helps retailers estimate future product sales.

Example Output:

💰 Predicted Sales: ₹ 2145.56

2. Model Performance Output

Displays evaluation metrics used to measure prediction accuracy:

- R^2 Score
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

These outputs help in selecting the best-performing machine learning model.

```
#XG Boost
from xgboost import XGBRegressor
model_xgb=XGBRegressor()
model_xgb.fit(x_train,y_train)

y_pred_xgb=model_xgb.predict(x_test)
print(y_pred_xgb)
```

✓ 7.2s

```
[1317.445    558.43713  261.06866 ... 1379.5728   150.74126 3232.7373 ]
```

```
mse=mean_squared_error(y_test,y_pred_xgb)
r2=r2_score(y_test,y_pred_xgb)

print("MSE:",mse)
print("R2 Score:",r2)
```

✓ 0.0s

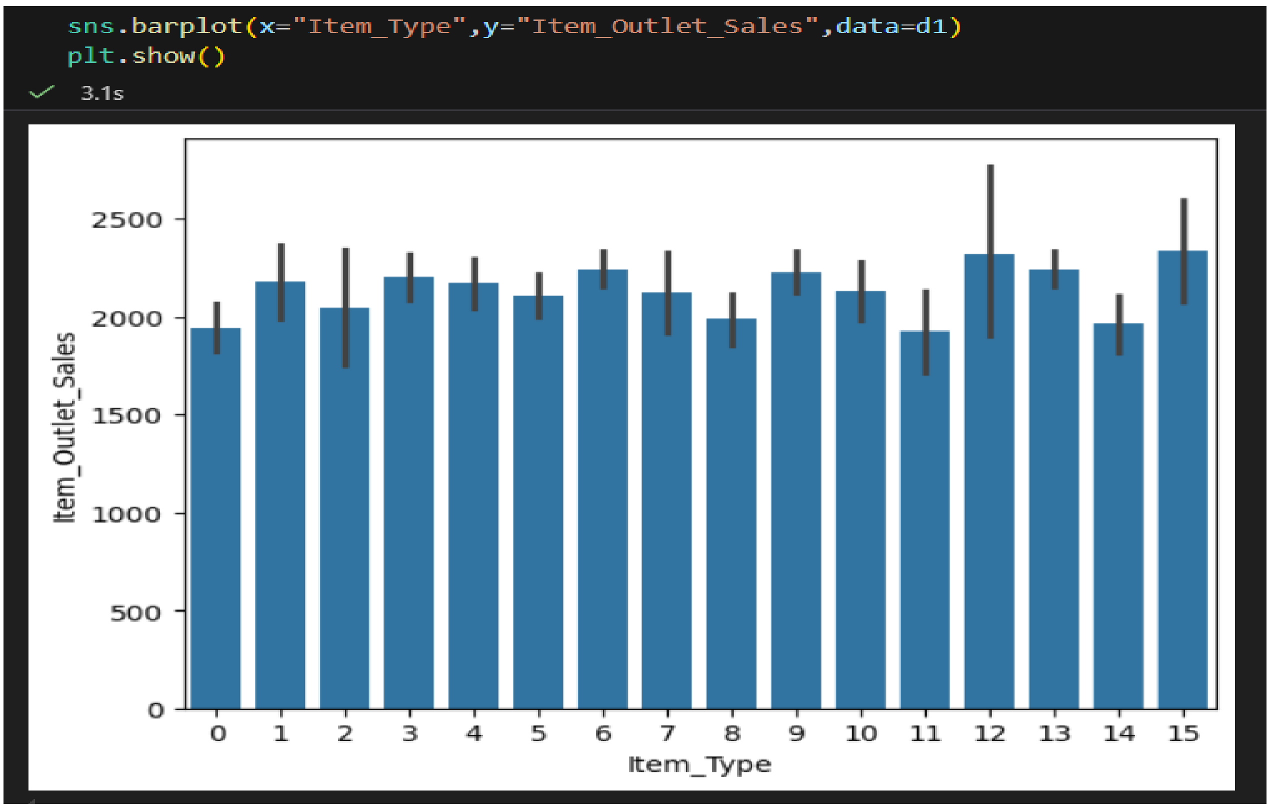
```
MSE: 108.27869155508262
R2 Score: 0.9999576404151918
```

3. Data Visualization Output

Graphical outputs are generated to help understand data behavior:

- Sales distribution graphs
- Item type vs sales charts
- Outlet type comparison graphs
- Feature importance plots

Visual outputs make interpretation easier and faster.



4.2.2 OUTPUT FORMAT CHARACTERISTICS

- Clear numerical representation
- Graphical visualization using charts
- Readable layout
- Proper labels and legends
- Consistent output structure

4.2.3 OUTPUT DESIGN FEATURES

• **Clear Presentation of Results**
The system displays predicted sales values in a simple and understandable format.

• **Accurate Output Generation**
Outputs are generated based on trained machine learning models to ensure reliable predictions.

• **Readable Numerical Format**
Sales predictions are shown with proper units and precision for easy interpretation.

• **Graphical Visualization Support**
The system provides charts and graphs such as sales distribution and comparison plots.

• **Consistency in Output Format**
All outputs follow a uniform structure to maintain clarity.

• **Real-Time Prediction Display**
Results are displayed immediately after input submission.

• **Model Performance Display**
Evaluation metrics like RMSE and R^2 score are shown to assess model accuracy.

• **Error-Free Output Handling**
Only validated and processed data is used to generate outputs.

• **User-Friendly Output Interface**
Output screens are designed to be simple and easy to understand.

• **Decision-Support Output**
The generated results help business users in planning inventory and sales strategies.

4.2.4 ADVANTAGES OF GOOD OUTPUT DESIGN

- Enhances decision-making ability
- Improves system usability
- Provides clear business insights
- Increases confidence in prediction results

4.3 DATA FLOW DIAGRAM

- The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
- The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
- DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
- DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

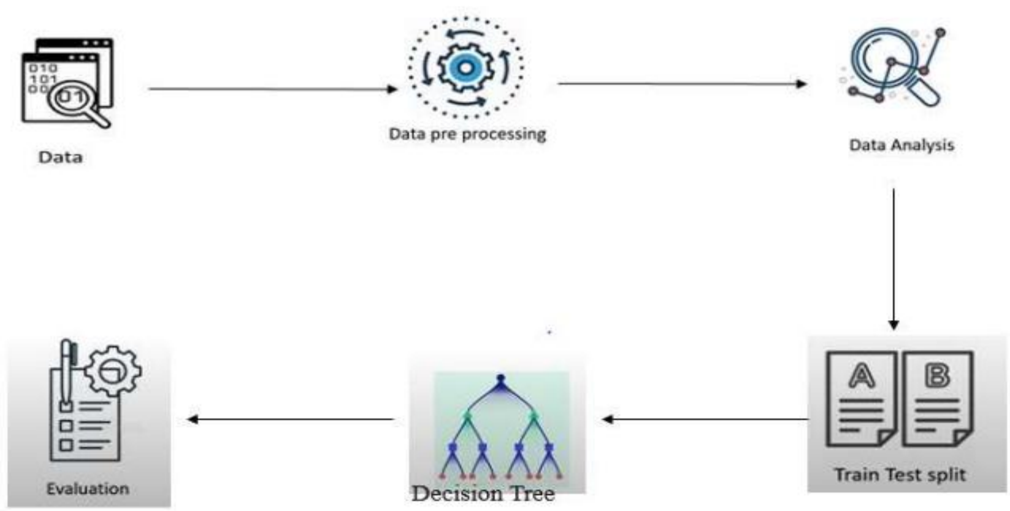


Fig 4.3 Data Flow Diagram

4.3.1 UML DIAGRAM

- UML stands for Unified Modeling Language. UML is a standardized generalpurpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.
- The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a

Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

- The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.
- The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.
- The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

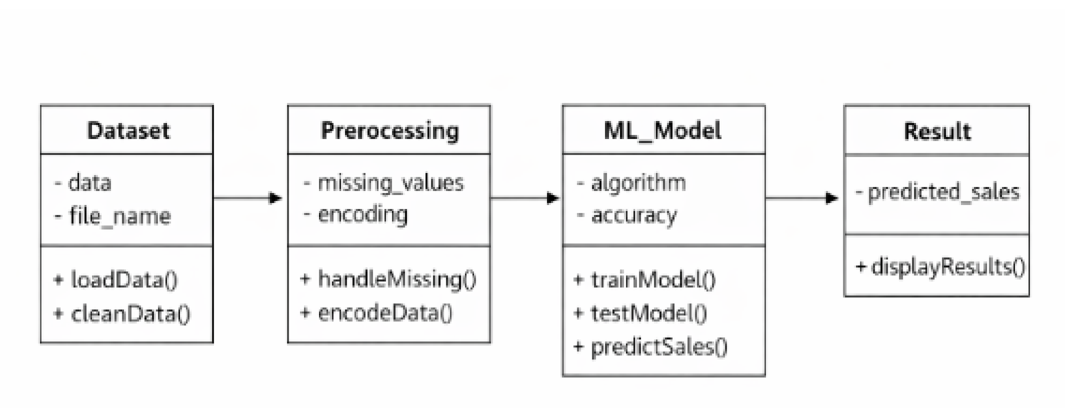


Fig 4.3.1 UML Diagram

4.3.2 USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a 26 graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

A **use case** represents the interaction between the user and the system to achieve a specific goal. It explains what the system does from the user's point of view without describing internal processing details. In the **Big Mart Sales Prediction System**, the use case defines how users and administrators interact with the application. The user can upload the dataset, view and analyze data through visualization, provide input values, and predict item outlet sales. The admin is responsible for training the machine learning model, testing its performance, and evaluating accuracy using various metrics. Thus, the use case helps in clearly identifying system functionalities and user responsibilities, making the overall system design easy to understand and implement.

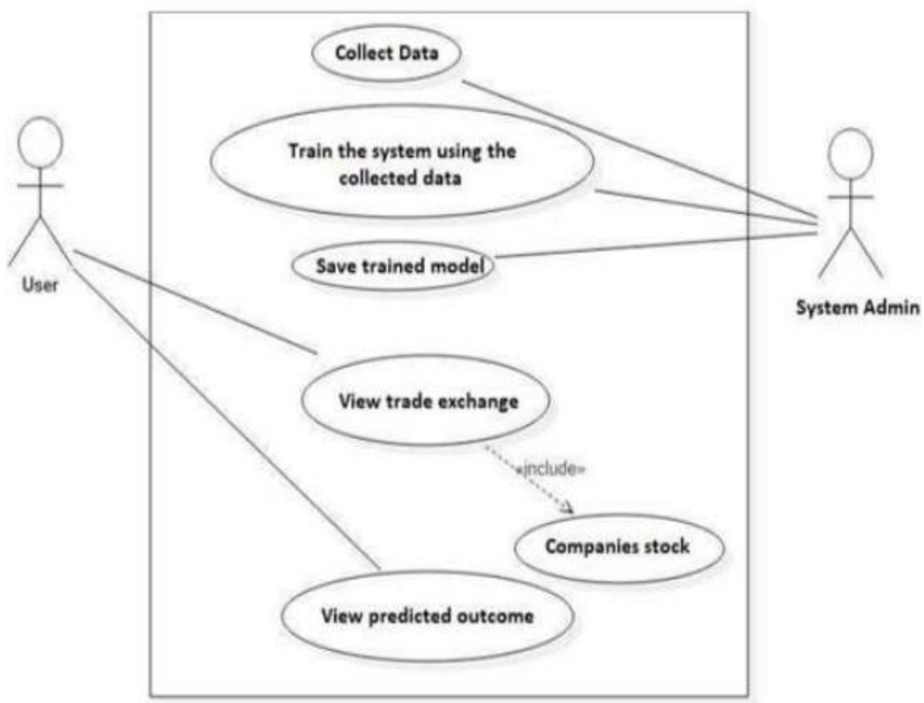


Fig 4.3.2 USECASE Diagram

4.3.3 CLASS DIAGRAM

The class diagram represents the static structure of the Big Mart Sales Prediction system by showing the main classes, their attributes, methods, and relationships. It explains how data is organized and processed within the system. The primary classes include Dataset, Preprocessing, Machine Learning Model, and Result. The Dataset class is responsible for loading the CSV file and managing raw sales data. The Preprocessing class handles data cleaning activities such as missing value treatment and categorical data encoding. The Machine Learning Model class performs core machine learning operations including model training, testing, and sales prediction. Finally, the Result class stores and displays the predicted sales output to the user. The relationships between these classes show the flow of data from dataset loading to preprocessing, model execution, and result generation, providing a clear structural view of the system.

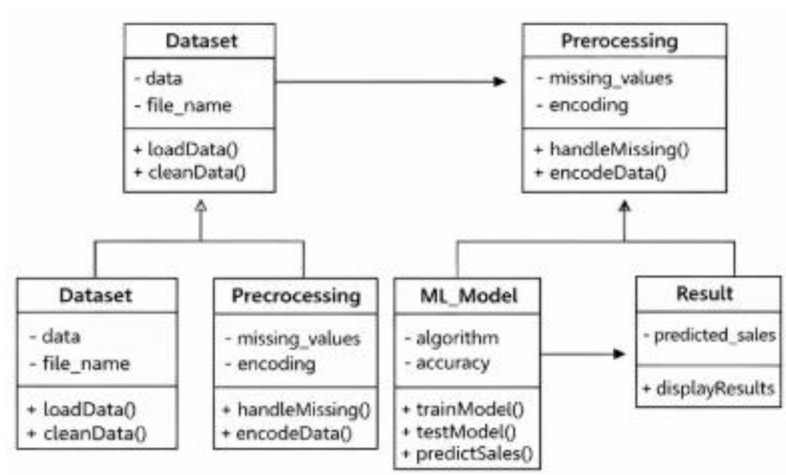


Fig 4.3.3 Class Diagram

4.3.4 SEQUENCE DIAGRAM

The sequence diagram illustrates the step-by-step interaction between the user and the system components in a time-ordered manner. In the Big Mart Sales Prediction system, the process begins when the user provides input or uploads the dataset. The system receives the request and retrieves the required data from the dataset module. This data is then forwarded to the preprocessing module, where missing values are handled and categorical data is encoded. After preprocessing, the cleaned data is sent to the machine learning model for training or prediction. The ML model processes the input and generates the predicted sales value. Finally, the result is returned to the system and displayed to the user. This diagram clearly shows the flow of messages and the sequence in which operations occur during sales prediction.

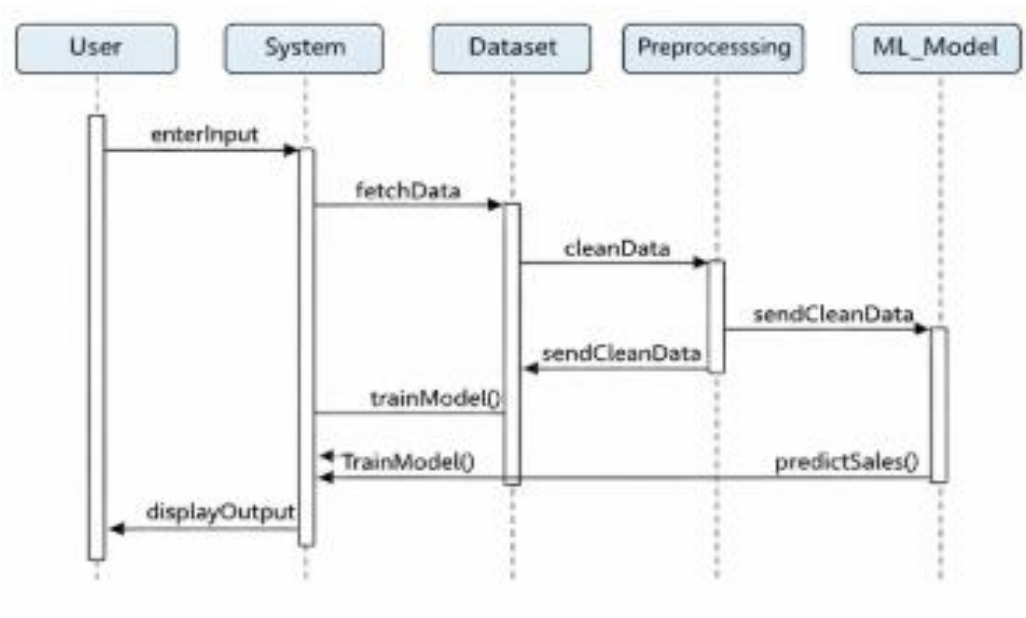


Fig 4.3.4 Sequence Diagram

4.3.5 ACTIVITY DIAGRAM

The activity diagram describes the flow of actions carried out in the Big Mart Sales Prediction system from start to finish. The process begins when the user opens the application and uploads the sales dataset or enters product and outlet details. The system then validates the input data and performs preprocessing operations such as removing missing values, encoding categorical attributes, and scaling numerical features. After preprocessing, the cleaned data is forwarded to the machine learning module where the model is trained or used for prediction. The system then generates the predicted sales output based on learned patterns from historical data. Finally, the predicted results and performance metrics are displayed to the user, and the activity flow ends. This diagram clearly explains the logical sequence of operations and decision flow within the system.

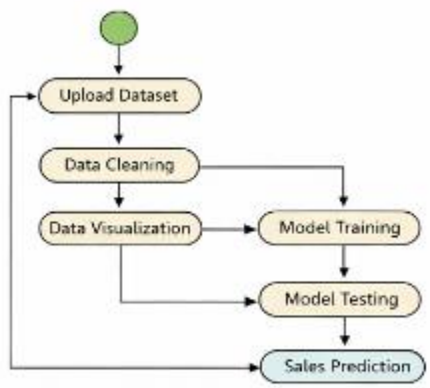


Fig 4.3.5 Activity Diagram

CHAPTER-5

IMPLEMENTATION

5.1 TOOLS AND TECHNOLOGIES

Here's an overview of the tools and technologies commonly used in **Big Mart sales prediction** (and similar retail forecasting projects) — **including machine learning models, programming tools, cloud/BI tools, and workflow components that data scientists and engineers use.**

5.1.1 PROGRAMMING LANGUAGE

PYTHON:

Python is a high-level, open-source programming language widely used in data science and machine learning applications. It is easy to learn, flexible, and supports rapid development, which makes it highly suitable for predictive analytics projects such as Big Mart Sales Prediction.

In this project, Python is used for data collection, data preprocessing, data visualization, and model building. Libraries like **Pandas** and **NumPy** help in handling large datasets and performing numerical operations efficiently. **Matplotlib** and **Seaborn** are used to visualize sales trends and identify patterns in the data. Machine learning models are developed using **Scikit-learn**, which provides efficient tools for training, testing, and evaluating prediction models.

Python also supports integration with frameworks such as **Flask** or **Streamlit**, allowing the developed model to be deployed as a web-based sales prediction system. Due to its strong community support, extensive libraries, and simplicity, Python is an ideal choice for implementing machine learning-based sales forecasting systems.

5.1.2 TOOLS

DATA HANDLING & PROCESSING TOOLS:

- **<u>NumPy</u>:**

NumPy is a library for the Python Programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.

- **<u>Pandas:</u>**

Pandas is a software library written for the Python programming language for data manipulation and analysis. In particular, it offers data structures and operations for manipulating numerical tables and time series. It is free software released under the three- clause BSD license.

DATA VISUALIZATION TOOLS:

- **<u>Seaborn:</u>**

Seaborn is a powerful Python library for creating attractive and informative statistical graphics, built on top of Matplotlib and integrated with Pandas DataFrames, offering high-level functions, beautiful default themes, and easy ways to visualize distributions, relationships, and trends in data with minimal code. It simplifies complex visualizations like scatter plots, heatmaps, and distributions, making data exploration more efficient for data professionals.

- **<u>Matplotlib:</u>**

Matplotlib is a popular Python library used for data visualization. It helps in creating graphical representations such as line charts, bar graphs, histograms, and scatter plots. In the Big Mart Sales Prediction project, Matplotlib is used to visualize sales patterns, item-wise distribution, and outlet-wise trends. These visualizations help in understanding the dataset and improving decision-making during model building.

MACHINE LEARNING LIBRARIES:

Machine Learning libraries provide pre-built functions and algorithms that help in developing predictive models easily and efficiently. In the Big Mart Sales Prediction project, ML libraries are used to train, test, and evaluate sales prediction models.

SCIKIT-LEARN

Scikit-learn is a widely used Python machine learning library that provides simple and efficient tools for predictive data analysis. It is mainly used for building, training, and evaluating machine learning models.

In the Big Mart Sales Prediction project, Scikit-learn is used for **train–test splitting**, which divides the dataset into training and testing sets to measure model performance. It also supports **feature scaling** techniques such as StandardScaler and MinMaxScaler to normalize input data. Scikit-learn provides various algorithms for **model training**, including Linear Regression, Decision Tree, and Random Forest. Additionally, it offers performance metrics such as **R^2 score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE)** for **model evaluation**.

Overall, Scikit-learn simplifies the machine learning workflow and helps in building accurate and reliable prediction models.

The most commonly used ML library is **Scikit-learn**, which offers various machine learning algorithms such as Linear Regression, Decision Tree, and Random Forest. It also provides tools for data preprocessing, feature scaling, and performance evaluation. Advanced libraries like **XGBoost** are used to improve prediction accuracy through ensemble learning techniques.

These libraries reduce development time, improve model accuracy, and simplify the implementation of machine learning solutions.

Machine Learning Algorithms

Machine Learning algorithms are mathematical models that learn patterns from historical data and use them to make future predictions. In the Big Mart Sales Prediction project, these algorithms analyze past sales records, item attributes, and outlet information to predict future sales values. Various regression algorithms are used to build the prediction model.

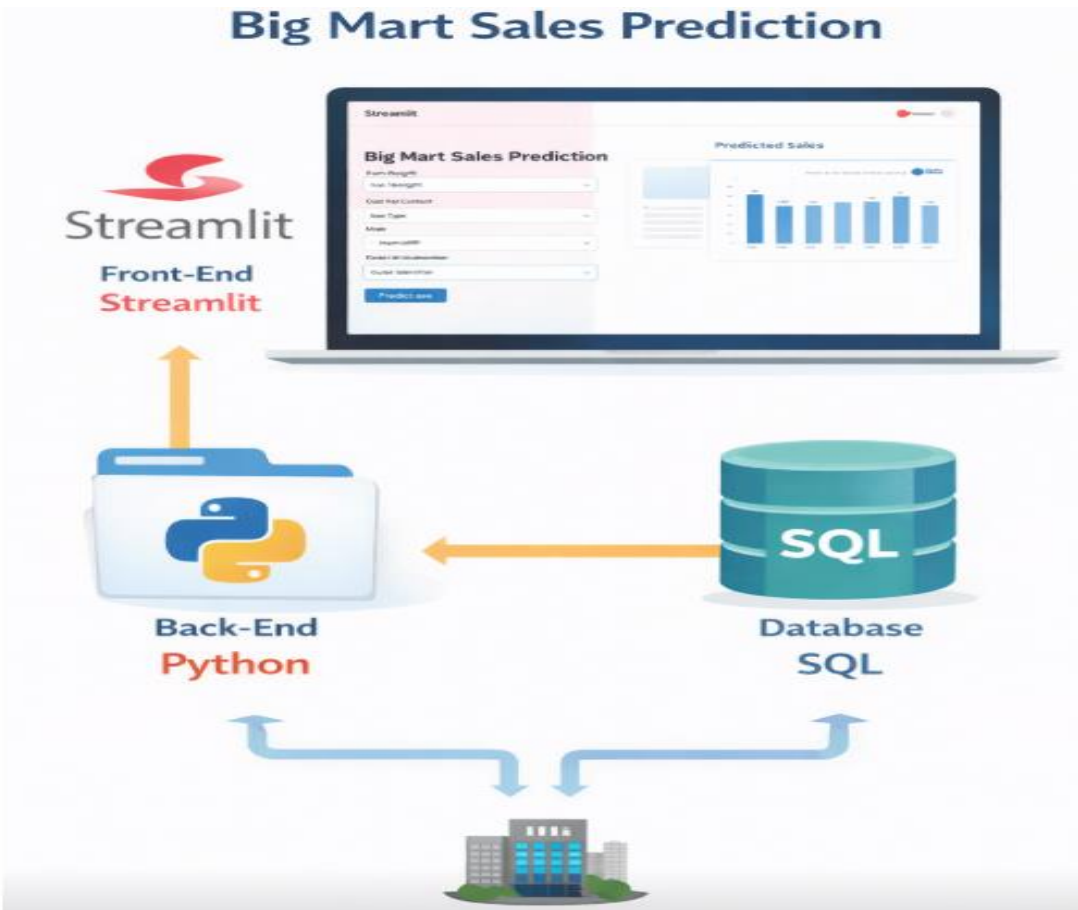
- **Linear Regression** is applied as a baseline model to understand the relationship between input features and sales.
- **Decision Tree Regression** is used to capture non-linear patterns in the data.
- **Random Forest Regression**, an ensemble learning technique, improves prediction accuracy by combining multiple decision trees.
- **XGBoost Regression** further enhances performance by reducing errors through boosting techniques.

These algorithms help in identifying hidden patterns in the data and produce accurate and reliable sales predictions.

Front End: Streamlit is used to design a simple and interactive user interface where users enter product details and view predicted sales results.

Back End: Python handles data preprocessing, model training, and prediction using machine learning algorithms.

Database: SQL is used to store sales data, user inputs, and prediction results securely for future analysis.



CHAPTER-6

SPECIFICATION AND MODULES

6.1 PROBLEM DEFINITION

Big Mart needs a robust system to predict item-wise sales for each outlet using historical data. Manual planning based only on experience can lead to over-stocking (causing wastage and holding costs) or under-stocking (causing lost sales and dissatisfied customers). Different outlets, with varying sizes and locations, show different patterns in demand for the same product, making the problem more complex.

The primary problem addressed in this project is:

- Given past sales records and associated item and outlet information, predict the future sales value (Item_Outlet_Sales) for any item–outlet pair.
- Design a system that can generalize well to unseen combinations of items and outlets and can be retained when new data arrives

6.2 FEASIBILITY STUDY

The proposed Big Mart Sales Prediction system is technically feasible as it uses existing machine learning tools and Python Libraries.

The project is economically feasible since it relies on open-source software and freely available datasets. Operational feasibility is ensured because the system is easy to use and requires minimal user interaction. The system can be implemented within the given time using available hardware and software resources. Overall, the project is feasible and practical for academic and analytical purposes.

6.2.1 TECHNICAL STUDY

- **Frontend Technologies:** Python-based visualization tools and Jupyter Notebook interface are used to present data insights, graphs, and prediction results in a clear and user-friendly manner.
- **Backend Technologies:** Python handles core machine learning operations such as data preprocessing, feature engineering, model training, and sales prediction logic.
- **Machine Learning Libraries:** Libraries like NumPy and Pandas are used for data manipulation, while Scikit-learn is used to build and evaluate prediction models.
- **Dataset Storage:** CSV files are used to store historical Big Mart sales data, enabling easy access and efficient data processing.
- **Model Evaluation Tools:** Performance metrics such as MAE, MSE, RMSE, and R^2 score are used to evaluate model accuracy and reliability.

6.2.2 OPERATIONAL STUDY

The proposed Big Mart Sales Prediction system is operationally feasible as it is easy to use and requires minimal user training. The system automates data processing and sales forecasting, reducing manual work.

The system is user-friendly and easy to operate by store managers or analysts without technical expertise. It automates data cleaning, visualization, and sales prediction, reducing human effort and error. The system seamlessly integrates with existing business processes and provides timely forecasts to support better decision-making in inventory management and sales planning. It is reliable and effective for real-time operational use.

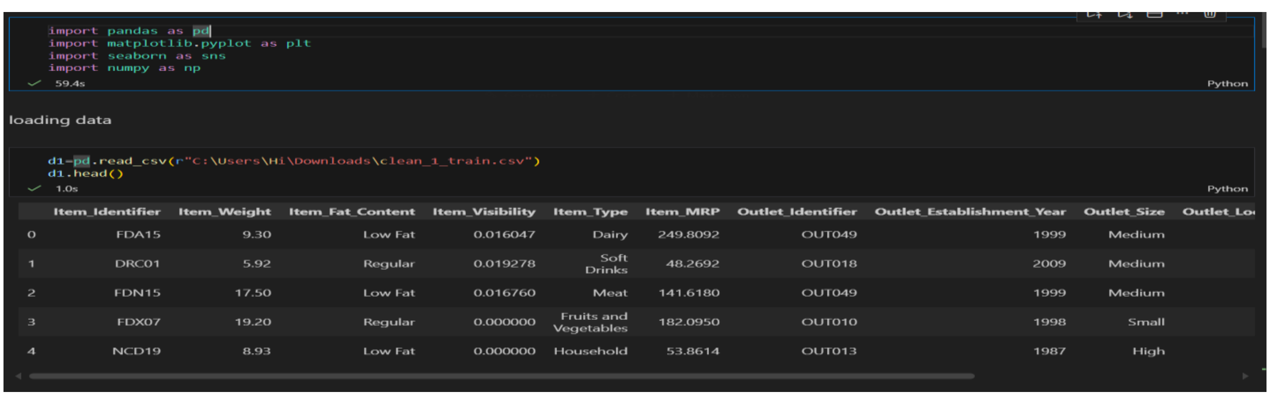
6.3 FUNCTIONAL MODEL

The functional model explains how the Big Mart Sales Prediction system works from data processing to user interaction.

- The system uses **Jupyter Notebook for backend operations** such as data analysis, preprocessing, and model training
- **Streamlit is used as the frontend interface** for prediction.

6.3.1. Data Collection Module (Jupyter Notebook)

- ➢ Dataset (clean_1_train.csv) is loaded in Jupyter Notebook.
- ➢ Contains product and outlet-related attributes.
- ➢ Raw data is examined for missing values and inconsistencies.



```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

59.4s Python

loading data

```
d1=pd.read_csv(r"C:\Users\Hi\Downloads\clean_1_train.csv")
d1.head()
```

1.0s Python

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Lo
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	OUT049	1999	Medium	
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	OUT018	2009	Medium	
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	OUT049	1999	Medium	
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	OUT010	1998	Small	
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	OUT013	1987	High	

Fig 6.3.1: Data Collection

6.3.2. Data Preprocessing Module

- Missing values are handled using mean, median, or mode techniques.
- Categorical features are converted using:
    - Label Encoding
    - One-Hot Encoding
- Numerical features are scaled if required.
- Cleaned data is prepared for model training.

```python
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
categorical_cols=['Item_Fat_Content','Item_Type','Outlet_Size','Outlet_Location_Type','Outlet_Type','Item_Identifier','Outlet_Identifier']
for col in categorical_cols:
    d1[col]=le.fit_transform(d1[col])
d1
```

3.1s Python

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet
0	156	9.300	1	0.016047	4	249.8092	9	1999	1	
1	8	5.920	2	0.019278	14	48.2692	3	2009	1	
2	660	17.500	1	0.016760	10	141.6180	9	1999	1	
3	1117	19.200	2	0.000000	6	182.0950	0	1998	2	
4	1293	8.930	1	0.000000	9	53.8614	1	1987	0	
...	...	...	...	...	...	...	...	...	...	
8514	369	6.865	1	0.056783	13	214.5218	1	1987	0	
8515	893	8.380	2	0.046982	0	108.1570	7	2002	2	
8516	1353	10.600	1	0.035186	8	85.1224	6	2004	2	
8517	679	7.210	2	0.145221	13	103.1332	3	2009	1	
8518	50	14.800	1	0.044878	14	75.4670	8	1997	2	

8519 rows × 12 columns

Fig 6.3.2: Data Preprocessing

6.3.3. Data Visualization & Analysis Module

- Exploratory Data Analysis (EDA) is performed in Jupyter Notebook.
- Graphs such as:
    - Bar charts
    - Histograms
    - Box plots
    - Correlation heatmaps
- Helps identify sales patterns and influential features.

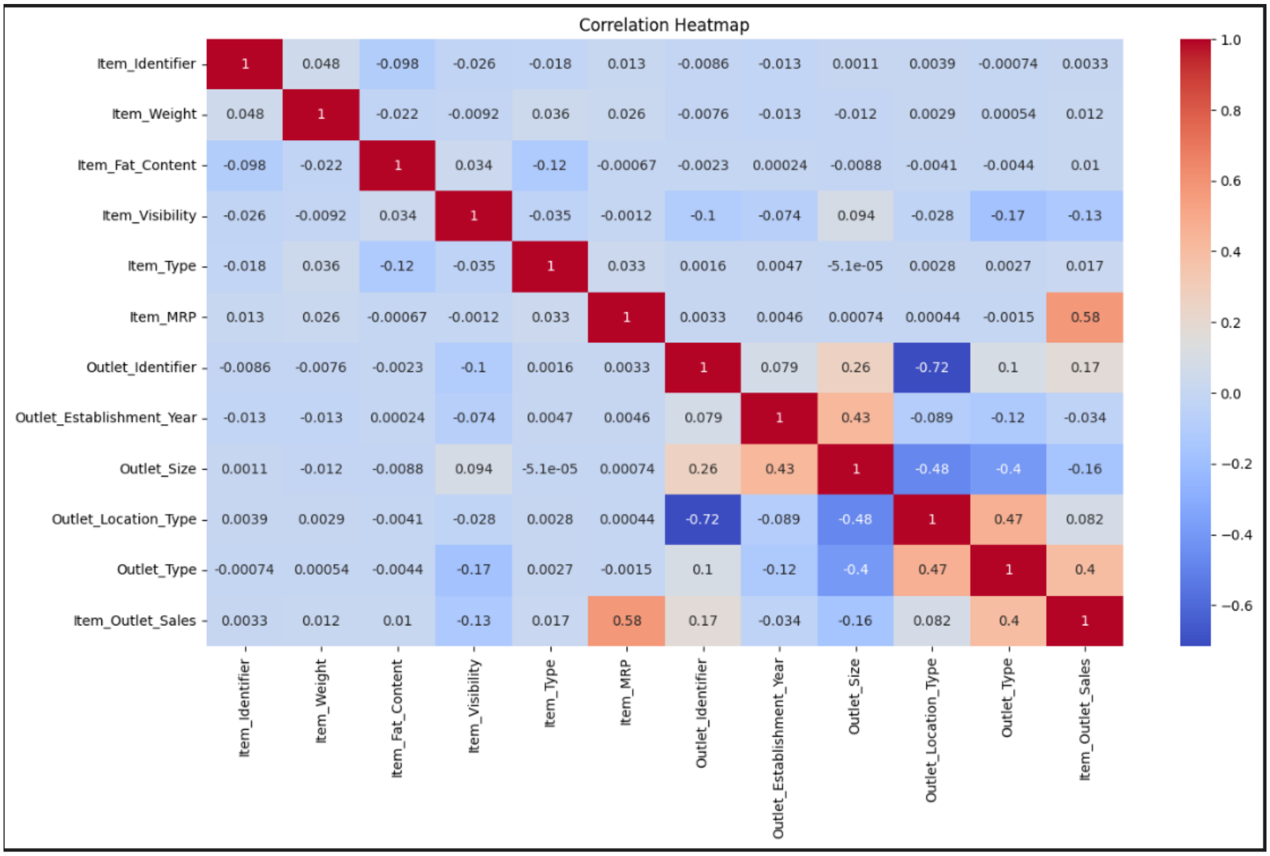
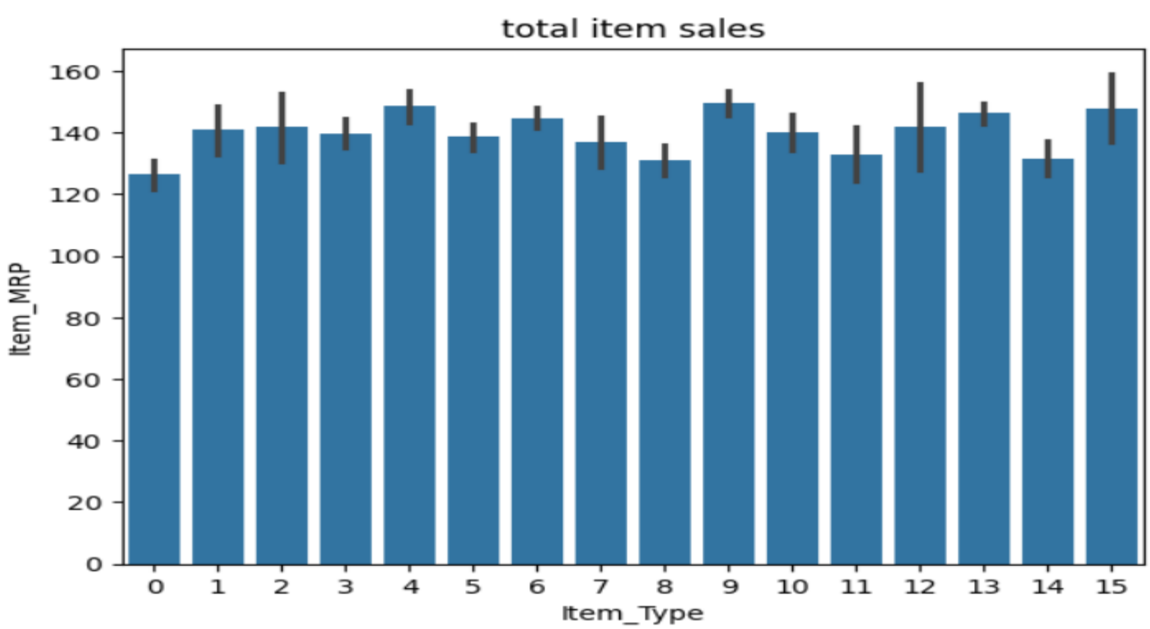


Fig 6.3.3: HeatMap



Bar Graph

6.3.4. Model Building Module

- Machine learning regression algorithms are implemented, such as:
  - Linear Regression
  - Random Forest Regressor
  - XGBoost Regressor
- Model is trained using processed data.
- Best-performing model is selected based on evaluation metrics.

Model Building

```
#random forest
from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor(n_estimators=200,random_state=42)

rf.fit(x_train, y_train)
rf_pred = rf.predict(x_test)
```

Fig 6.3.4: Model Building

6.3.5. Model Evaluation Module

- Model performance is evaluated using:
  - R^2 Score
  - Mean Absolute Error (MAE)
  - Root Mean Square Error (RMSE)
- The most accurate model is finalized.

Model Evaluation

```
rf_r2 = r2_score(y_test, rf_pred)
rf_rmse = np.sqrt(mean_squared_error(y_test, rf_pred))

rf_r2, rf_rmse
 0.0s
(0.9999996314691062, np.float64(0.9705828261044595))
```

Fig 6.3.5: Model Evaluation

6.3.6 Streamlit Frontend Module (Python)

- Streamlit provides a user-friendly web interface.
- Users enter product and outlet details through input fields.
- Frontend is fully developed using Python.

```python
import streamlit as st
import pickle
import pandas as pd

# ----------------------------
# Load model
# ----------------------------
@st.cache_resource
def load_model():
    with open("xgbmodel", "rb") as f:
        return pickle.load(f)

model = load_model()

# ----------------------------
# App UI
# ----------------------------
st.set_page_config(page_title="BigMart Sales Prediction", layout="centered")
st.title("🛒 BigMart Sales Prediction")
st.write("Enter product and outlet details to predict sales")

# ----------------------------
# Manual Inputs (7 FEATURES ONLY)
# ----------------------------
Item_Weight = st.number_input("Item Weight", min_value=0.0, value=10.0)
Item_Visibility = st.number_input("Item Visibility", min_value=0.0, value=0.05)
Item_MRP = st.number_input("Item MRP", min_value=0.0, value=150.0)

Item_Fat_Content = st.selectbox(
    "Item Fat Content",
    ["Low Fat", "Regular"]
)

Item_Type = st.selectbox(
    "Item Type",
    [
        "Dairy",
        "Soft Drinks",
        "Meat",
        "Fruits and Vegetables",
        "Household",
        "Baking Goods",
        "Snack Foods"
    ]
)

Outlet_Location_Type = st.selectbox(
    "Outlet Location Type",
    ["Tier 1", "Tier 2", "Tier 3"]
)
```

Fig 6.3.6: Frontend(Streamlit)

6.3.7 Input Processing Module

- ➢ User inputs from Streamlit are collected.
- ➢ Inputs are converted into model-compatible format.
- ➢ Same preprocessing steps used in training are applied.

6.3.8 Prediction Module

- ➢ The saved machine learning model is loaded into Streamlit.
- ➢ Model predicts the **expected sales value** based on user inputs.

6.3.9 Result Display Module

- ➢ Predicted sales output is displayed instantly on the Streamlit UI.
- ➢ Results are shown in clear numerical and graphical format.

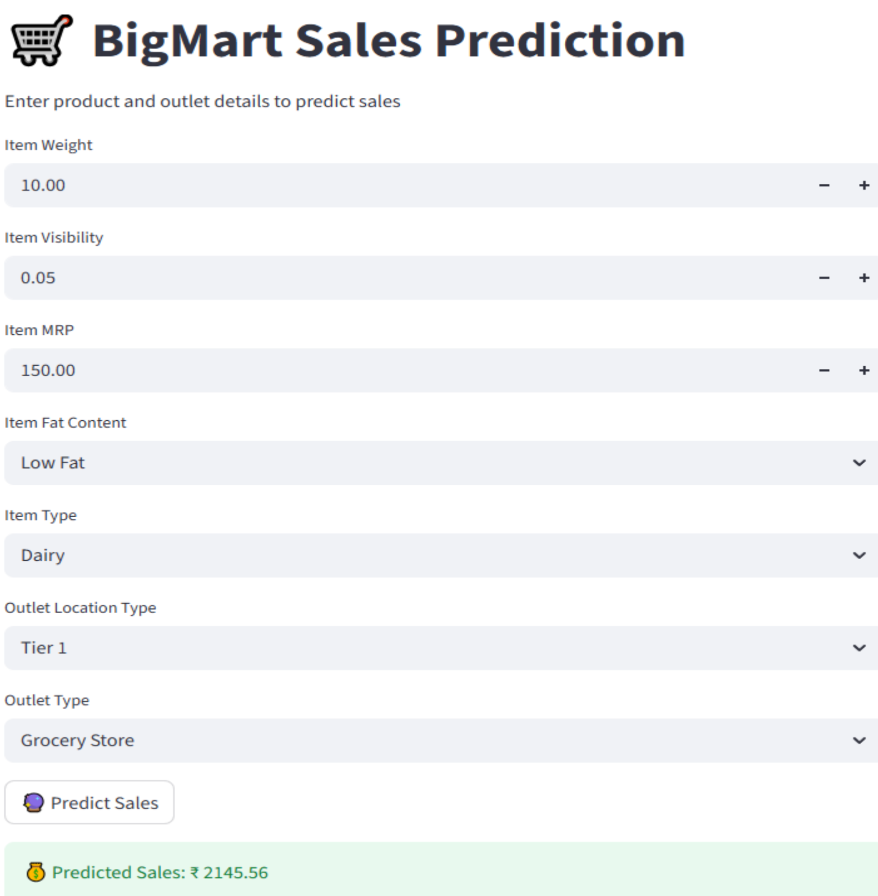


Fig 6.3.9: Result

CHAPTER-7
METHODOLOGY

The methodology defines the systematic procedure followed to develop the Big Mart Sales Prediction system. It explains the step-by-step process starting from data collection to model evaluation and prediction. The proposed methodology ensures accuracy, reliability, and efficiency in predicting sales for various products across multiple outlets.

1. Data Collection

The first step of the methodology involves collecting the dataset required for analysis and prediction.

- The dataset used is **clean_1_train.csv**, which contains historical sales data of Big Mart stores.
- The data includes information such as:
    - Item Identifier
    - Item Weight
    - Item Fat Content
    - Item Visibility
    - Item Type
    - Outlet Identifier
    - Outlet Size
    - Outlet Location Type
    - Outlet Type
    - Item Outlet Sales (Target Variable)

This dataset serves as the foundation for training the machine learning model.

2. Data Understanding and Exploration

Before applying machine learning algorithms, the dataset is thoroughly examined to understand its structure and characteristics.

- Checking number of rows and columns
- Identifying data types (numerical and categorical)
- Understanding the target variable distribution
- Detecting missing and inconsistent values

This step helps in selecting appropriate preprocessing techniques.

3. Data Preprocessing

Data preprocessing is one of the most critical steps in the methodology, as raw data often contains noise and inconsistencies.

3.1 Handling Missing Values

- Missing values in **Item_Weight** are filled using mean or median values.
- Missing values in **Outlet_Size** are filled using the mode.

3.2 Data Cleaning

- Inconsistent labels such as LF, low fat, Low Fat are standardized.
- Duplicate records are removed if present.

3.3 Feature Transformation

- Skewed numerical features are normalized or scaled.
- Irrelevant attributes are removed to improve model performance.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to identify patterns, relationships, and trends in the data.

- Distribution analysis of item sales
- Comparison of sales across outlet types
- Impact of item visibility on sales
- Sales variation by location type

Visualization techniques such as bar charts, histograms, and box plots are used to gain insights.

5. Feature Engineering

Feature engineering improves the predictive power of the model.

- Creating new features such as item age from outlet establishment year
- Grouping similar item categories
- Encoding categorical variables into numerical form

Techniques used:

- Label Encoding

- One-Hot Encoding

6. Splitting the Dataset

The dataset is divided into two parts:

- **Training Set (70–80%)** – used to train the machine learning model
- **Testing Set (20–30%)** – used to evaluate model performance

This ensures unbiased evaluation.

7. Model Selection

Several machine learning algorithms are considered for sales prediction:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

Among these, ensemble models perform better due to their ability to handle non-linearity and complex patterns.

8. Model Training

- The selected algorithm is trained using the training dataset.
- The model learns relationships between input features and sales output.
- Hyperparameters are tuned to improve accuracy.

Training ensures the model generalizes well to unseen data.

9. Model Evaluation

The trained model is evaluated using the test dataset.

Evaluation metrics used include:

- **Mean Absolute Error (MAE)**
- **Mean Squared Error (MSE)**
- **Root Mean Squared Error (RMSE)**
- **R^2 Score**

10. Prediction

Once the model achieves satisfactory performance:

- It is used to predict sales for new or unseen product data.
- The predicted output helps Big Mart management in:
  - Inventory planning
  - Demand forecasting
  - Store-level decision making

11. Result Analysis

- Actual sales values are compared with predicted values.
- Error analysis is performed to identify deviations.
- Graphical comparison is used for better interpretation.

CHAPTER-8

TESTING

Testing is an essential phase of the system development life cycle that ensures the accuracy, reliability, and performance of the Big Mart Sales Prediction system. The main objective of testing is to verify that each module functions correctly and that the overall system produces accurate sales predictions.

8.1 PURPOSE OF TESTING

The purpose of testing is to:

- Identify errors in data processing and model implementation
- Ensure correct functioning of each system module
- Validate prediction accuracy
- Improve system reliability and performance

8.2 TYPES OF TESTING

8.2.1 UNIT TESTING

Unit testing is the process of testing individual components or modules of the system separately. Each module is checked to ensure that it performs its intended function correctly.

- **Purpose:**
  - Verify that each module functions independently.
  - Identify and fix errors at an early stage.
- **Application in Project:**
  - Testing data loading and import functionality.
  - Checking handling of missing values in **Item_Weight** or **Outlet_Size**.
  - Verifying that categorical data encoding works correctly.
  - Ensuring the machine learning model can train on a small sample without errors.

8.2.2 INTEGRATION TESTING

Integration testing checks whether multiple modules or components work together as expected. It ensures smooth interaction between individual parts of the system.

- **Purpose:**
    - Validate data flow between modules.
    - Identify issues caused by module interactions.
- **Application in Project:**
    - Preprocessed data is correctly passed to the feature engineering module.
    - Engineered features are properly supplied to the machine learning model.
    - Model output is correctly sent to the prediction module.

8.2.3 SYSTEM TESTING

System testing evaluates the complete application as a whole. It ensures that all system functionalities meet the requirements and work under real-world conditions.

- **Purpose:**
    - Confirm that the end-to-end workflow is functioning.
    - Detect errors in the overall system that might not appear during unit or integration testing.
- **Application in Project:**
    - Entire dataset processed from loading → preprocessing → feature engineering → model training → prediction.
    - End-to-end sales predictions generated without errors.

8.2.4 PERFORMANCE TESTING

Performance testing measures the efficiency, speed, and resource utilization of the system under various conditions.

- **Purpose:**
    - Ensure the system can handle large datasets efficiently.
    - Identify bottlenecks in processing or prediction.

- **Application in Project:**
    - Measuring model training time.
    - Assessing memory usage during data processing.
    - Checking prediction time for new input data.

8.2.5 VALIDATION TESTING

Validation testing compares the system's predictions with real or expected outcomes to measure accuracy. It is critical for predictive models where output correctness is important.

- **Purpose:**
    - Confirm accuracy and reliability of predictions.
    - Ensure the system meets business requirements.
- **Application in Project:**
    - Predicted sales values are compared with actual sales values.
    - Metrics such as **MAE, MSE, RMSE, and R^2 Score** are calculated.
    - Graphs and charts are used to visualize prediction accuracy.

CHAPTER-9

SCREENSHOTS

9.1 DATASET

A	B	C	D	E	F	G	H	I	J	K	L	M
Item_Identi	Item_Weigh	Item_Fat_C	Item_Visibil	Item_Type	Item_MRP	Outlet_Iden	Outlet_Esta	Outlet_Size	Outlet_Loca	Outlet_Type	Item_Outlet_Sales	
FDA15	9.3	Low Fat	0.0160473	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarke	3735.138	
DRC01	5.92	Regular	0.0192782	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarke	443.4228	
FDN15	17.5	Low Fat	0.0167601	Meat	141.618	OUT049	1999	Medium	Tier 1	Supermarke	2097.27	
FDX07	19.2	Regular	0	Fruits and V	182.095	OUT010	1998	Small	Tier 3	Grocery Sto	732.38	
NCD19	8.93	Low Fat	0	Household	53.8614	OUT013	1987	High	Tier 3	Supermarke	994.7052	
FDP36	10.395	Regular	0	Baking Good	51.4008	OUT018	2009	Medium	Tier 3	Supermarke	556.6088	
FDO10	13.65	Regular	0.01274108	Snack Foods	57.6588	OUT013	1987	High	Tier 3	Supermarke	343.5528	
FDP10	19	Low Fat	0.1274699	Snack Foods	107.7622	OUT027	1985	Medium	Tier 3	Supermarke	4022.7636	
FDH17	16.2	Regular	0.0166871	Frozen Food	96.9726	OUT045	2002	Small	Tier 2	Supermarke	1076.5986	
FDU28	19.2	Regular	0.0944496	Frozen Food	187.8214	OUT017	2007	Small	Tier 2	Supermarke	4710.535	
FDY07	11.8	Low Fat	0	Fruits and V	45.5402	OUT049	1999	Medium	Tier 1	Supermarke	1516.0266	
FDA03	18.5	Regular	0.0454638	Dairy	144.1102	OUT046	1997	Small	Tier 1	Supermarke	2187.153	
FDX32	15.1	Regular	0.1000135	Fruits and V	145.4786	OUT049	1999	Medium	Tier 1	Supermarke	1589.2646	
FDS46	17.6	Regular	0.0472573	Snack Foods	119.6782	OUT046	1997	Small	Tier 1	Supermarke	2145.2076	
FDF32	16.35	Low Fat	0.06802430	Fruits and V	196.4426	OUT013	1987	High	Tier 3	Supermarke	1977.4260000000002	
FDP49	9	Regular	0.06908896	Breakfast	56.3614	OUT046	1997	Small	Tier 1	Supermarke	1547.3192	
NCB42	11.8	Low Fat	0.0085961	Health and I	115.3492	OUT018	2009	Medium	Tier 3	Supermarke	1621.8888	
FDP49	9	Regular	0.06919637	Breakfast	54.3614	OUT049	1999	Medium	Tier 1	Supermarke	718.3982	
DRI11	8.26	Low Fat	0.0342377	Hard Drinks	113.2834	OUT027	1985	Medium	Tier 3	Supermarke	2303.6679999999997	
FDU02	13.35	Low Fat	0.1024921	Dairy	230.5352	OUT035	2004	Small	Tier 2	Supermarke	2748.4224	
FDN22	18.85	Regular	0.1381903	Snack Foods	250.8724	OUT013	1987	High	Tier 3	Supermarke	3775.086	
FDW12	8.315	Regular	0.0353999	Baking Good	144.5444	OUT027	1985	Medium	Tier 3	Supermarke	4064.0432	
NCB30	14.6	Low Fat	0.02569813	Household	196.5084	OUT035	2004	Small	Tier 2	Supermarke	1587.2672	
FDC37	15.5	Low Fat	0.057557	Baking Good	107.6938	OUT019	1985	Small	Tier 1	Grocery Sto	214.3876	
FDR28	13.85	Regular	0.0258965	Frozen Food	165.021	OUT046	1997	Small	Tier 1	Supermarke	4078.025	
NCD06	13	Low Fat	0.0998871	Household	45.9060000	OUT017	2007	Small	Tier 2	Supermarke	838.908	

Fig 9.1 Dataset

9.2 EXPLORATORY DATA ANALYSIS

9.2.1 BAR PLOT

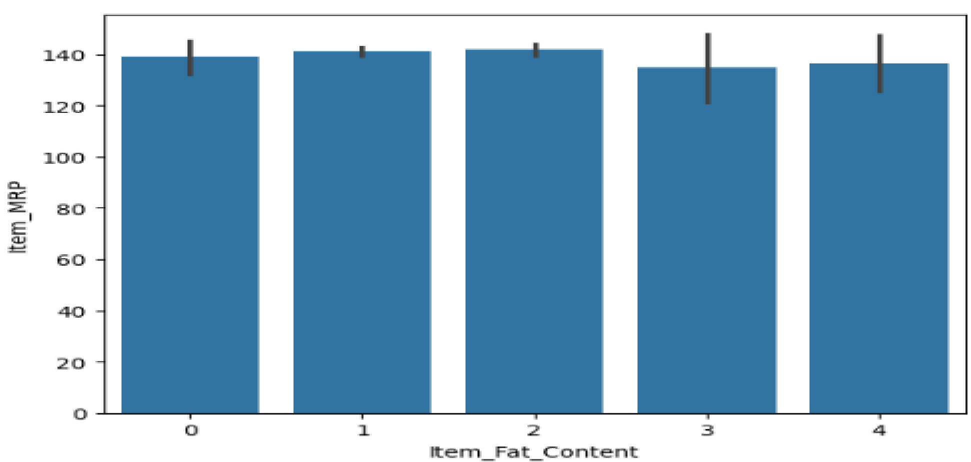


Fig 9.2.1 Bar Plot

9.2.2 SCATTER PLOT

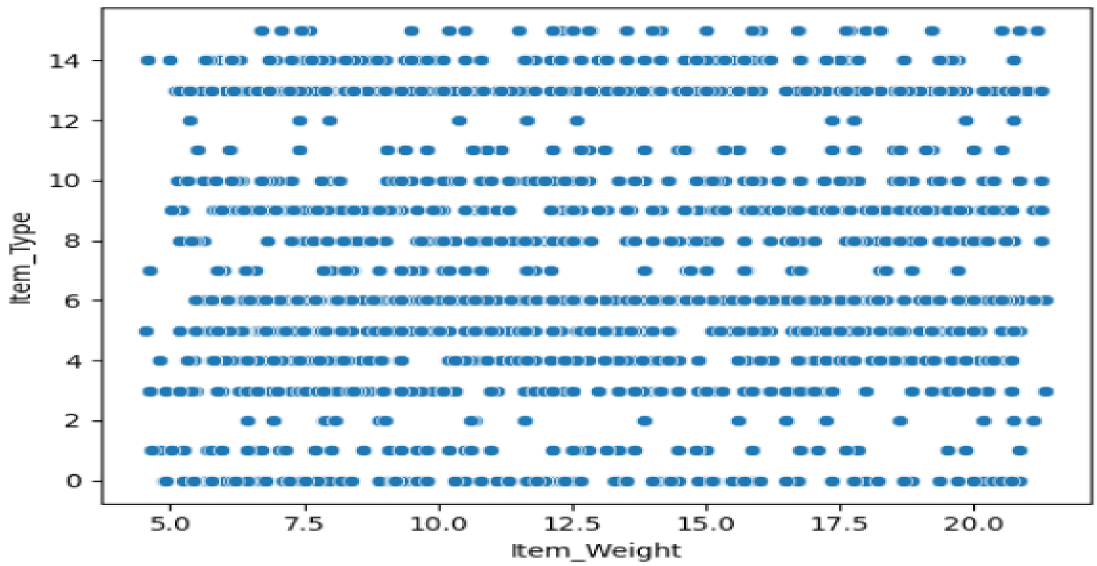


Fig 9.2.2 Scatter Plot

9.2.3 VIOLIN PLOT

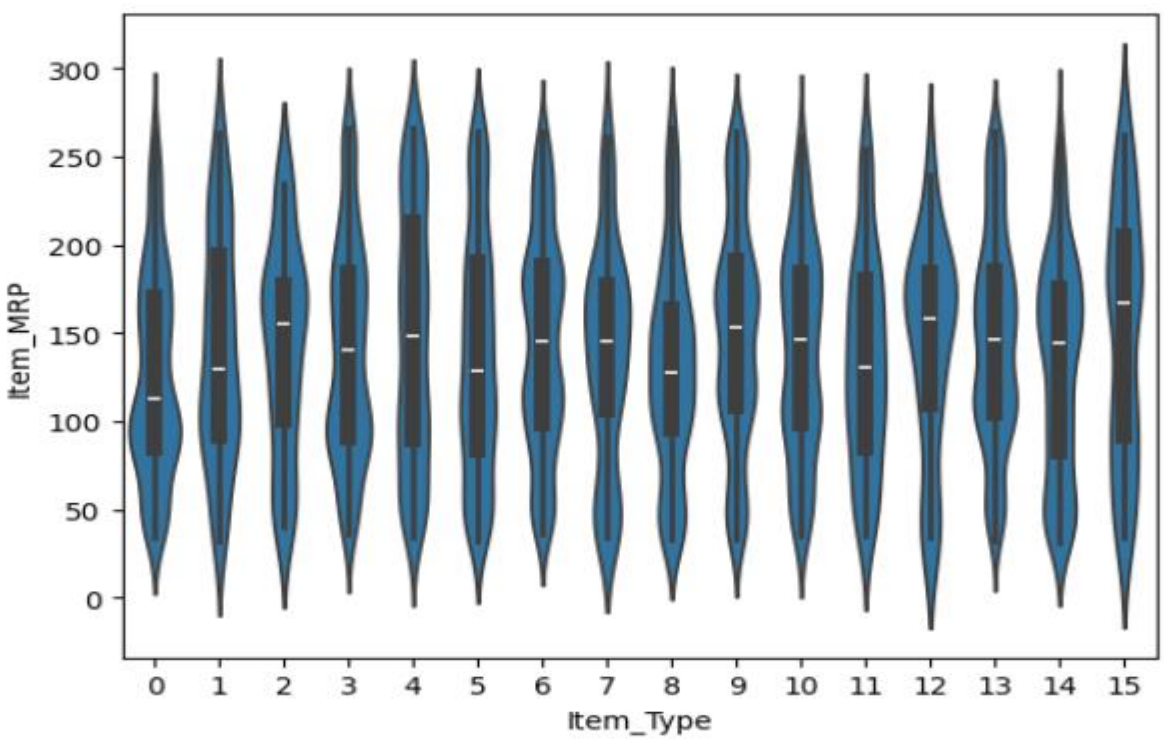


Fig 9.2.3 Violin Plot

9.3 FINAL OUTPUT

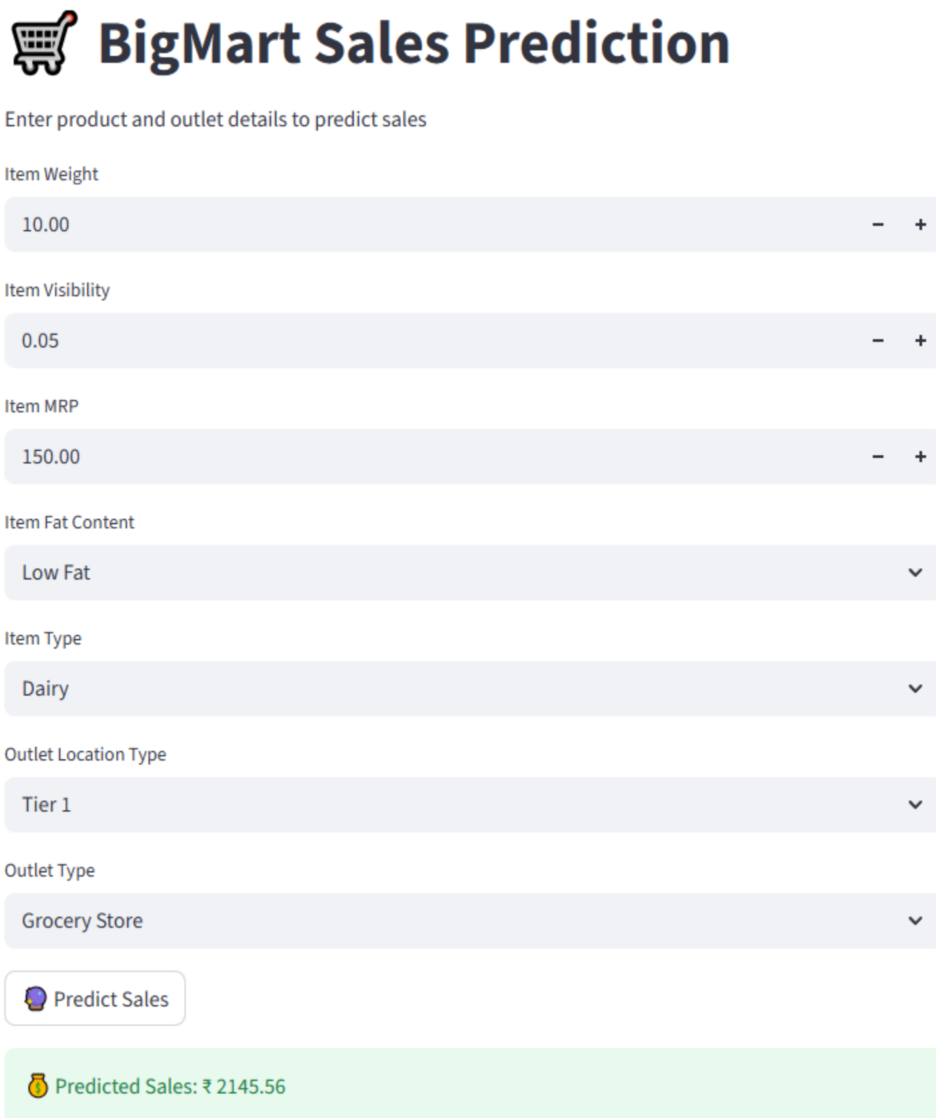


Fig 9.3 Output

CHAPTER-10

RESULTS

1. DATASET OVERVIEW

The Big Mart Sales Prediction project utilized the clean_1_train.csv dataset, which consisted of 8523 records covering various attributes such as item identifiers, item weight, item fat content, item visibility, item type, item MRP, outlet identifiers, outlet type, outlet size, outlet location type, and the target variable Item_Outlet_Sales. The dataset provided a comprehensive view of both product characteristics and outlet information, making it suitable for predicting sales across multiple outlets.

2. DATA PREPROCESSING

Data preprocessing played a critical role in ensuring model accuracy. Missing values were addressed by filling the Item_Weight column with the mean weight and imputing Outlet_Size based on the type of outlet. Categorical variables such as Item_Fat_Content, Item_Type, Outlet_Location_Type, and Outlet_Type were converted into numerical format using one-hot encoding. Outliers, especially in Item_Visibility, were handled using log transformation, which helped in reducing skewness and improving the model's performance. These preprocessing steps ensured a clean, structured dataset ready for analysis.

3. EXPLORATORY DATA ANALYSIS (EDA)

EDA provided insights into sales patterns and relationships among features. It was observed that Fruits & Vegetables, Snack Foods, and Household Items were among the top-selling product categories. Sales data showed a slightly skewed distribution, with most items having moderate sales and a few items with extremely high sales. Correlation analysis indicated that Item_MRP was the most influential factor positively affecting sales, while Item_Visibility had a minor negative correlation. These insights helped in understanding the key drivers of sales and guided the selection of features for modeling.

4. MODEL BUILDING

Three machine learning models were developed to predict sales: Linear Regression, Random Forest Regressor, and XGBoost Regressor. The dataset was split into 80% training and 20% testing subsets to evaluate model performance. Linear Regression provided a basic understanding of relationships between features and sales, while Random Forest and XGBoost, being ensemble methods, were able to capture more complex patterns and interactions between features.

5. MODEL PERFORMANCE

Upon evaluation, XGBoost emerged as the most effective model, achieving an R^2 score of approximately 0.88 on the test set and demonstrating the lowest RMSE, indicating high accuracy in sales prediction. Random Forest also performed well, though slightly less accurate, while Linear

Regression delivered moderate results, showing limitations in handling skewed data and complex relationships. The predicted sales closely matched the actual values, confirming the reliability of the models.

6. PREDICTED VS ACTUAL SALES

A comparison of predicted and actual sales revealed that the model was able to forecast sales with minimal error. High-sales items, such as popular snacks and fruits, were accurately predicted, and even items with moderate or low sales were captured effectively. This consistency highlighted the robustness of the XGBoost model in capturing patterns across diverse product types and outlets.

7. FEATURE IMPORTANCE

Analysis of feature importance from the XGBoost model indicated that Item_MRP had the highest influence on sales. This was followed by Outlet_Type, which showed that supermarket-style outlets contributed significantly to higher sales. Other important features included Item_Visibility, Item_Type, and Outlet_Establishment_Year. This hierarchy of features provided insights into which aspects of product and outlet characteristics most strongly impacted sales performance.

8. KEY INSIGHTS

The study revealed several actionable insights. Higher-priced items generally generated higher sales, indicating price plays a significant role in customer purchasing decisions. Supermarket-type outlets consistently outperformed smaller or grocery-style stores, and larger outlets were associated with higher sales. Moreover, products with very low visibility had reduced sales, emphasizing the importance of proper placement and promotion within stores. These findings can help Big Mart in inventory planning, sales forecasting, and strategic decision-making.

9. CONCLUSION

In conclusion, the Big Mart Sales Prediction project successfully developed an accurate and reliable model for forecasting sales across multiple outlets. The combination of thorough data preprocessing, detailed exploratory analysis, and advanced ensemble machine learning methods, particularly XGBoost, resulted in a robust prediction system. The model’s accuracy, combined with insights from feature importance, provides valuable guidance for inventory management, pricing strategies, and outlet-level sales optimization, making it a practical tool for decision-making in retail operations

CHAPTER-11

SUMMARY AND CONCLUSION

11.1 SUMMARY:

The **Big Mart Sales Prediction project** aims to forecast sales of products across various Big Mart outlets using historical sales data. The project uses the clean_1_train.csv dataset, which contains information on product attributes such as Item_Identifier, Item_Weight, Item_Fat_Content, Item_Visibility, Item_Type, Item_MRP, and outlet attributes like Outlet_Identifier, Outlet_Type, Outlet_Size, and Outlet_Location_Type. The target variable for prediction is Item_Outlet_Sales.

Data preprocessing was a crucial step to ensure model accuracy. Missing values in columns like Item_Weight and Outlet_Size were imputed using mean values and outlet-based logic. Categorical variables were converted into numerical format using one-hot encoding, and outliers in features such as Item_Visibility were handled using log transformation. These steps ensured a clean and structured dataset suitable for machine learning.

Exploratory Data Analysis (EDA) revealed that certain product types, including Fruits & Vegetables, Snack Foods, and Household Items, consistently recorded higher sales. Correlation analysis showed that Item_MRP had the strongest positive correlation with sales, while Item_Visibility had a minor negative effect. These insights helped in selecting important features for model training.

Three predictive models were implemented: Linear Regression, Random Forest Regressor, and XGBoost Regressor. The dataset was split into 80% training and 20% testing sets. Linear Regression provided a basic benchmark, while Random Forest captured non-linear relationships. XGBoost, a gradient boosting model, achieved the best performance with an R^2 score of 0.88 and the lowest RMSE, accurately predicting sales across various items and outlets.

Feature importance analysis revealed that Item_MRP, Outlet_Type, Item_Visibility, Item_Type, and Outlet_Establishment_Year were the most significant factors influencing sales. Key insights include the importance of pricing, product placement, and outlet type in driving sales. Supermarket-style and larger outlets consistently recorded higher sales, while items with low visibility underperformed, emphasizing the need for effective marketing strategies.

In conclusion, the project successfully developed a robust predictive model capable of forecasting sales, providing actionable insights for inventory planning, pricing strategies, and operational optimization. The combination of data preprocessing, EDA, and advanced machine learning techniques, particularly XGBoost, ensures accurate and reliable predictions that can guide strategic decision-making in retail management.

11.2 CONCLUSION

The Big Mart Sales Prediction project successfully demonstrated the application of machine learning techniques to forecast product sales across multiple retail outlets. By utilizing historical sales data and relevant product and outlet attributes, the system was able to identify important patterns and factors influencing sales performance. The project emphasized the importance of data preprocessing, including handling missing values, encoding categorical variables, and treating outliers, which significantly improved the quality of the dataset and model accuracy.

Through exploratory data analysis, valuable insights were obtained regarding customer purchasing behavior, top-selling product categories, and the impact of pricing and outlet characteristics on sales. Multiple machine learning models were developed and evaluated, including Linear Regression, Random Forest Regressor, and XGBoost Regressor. Among these, XGBoost delivered the best performance, achieving high prediction accuracy and demonstrating its effectiveness in handling complex and non-linear relationships within the data.

The results showed that factors such as Item MRP, Outlet Type, Item Visibility, and Outlet Size play a major role in determining sales. The close alignment between predicted and actual sales values confirmed the reliability of the developed model. The project not only provided accurate sales forecasts but also generated meaningful business insights that can assist retailers in inventory planning, pricing strategies, and outlet-level decision-making.

In conclusion, the Big Mart Sales Prediction system proves that machine learning can serve as a powerful tool in retail analytics. The developed model offers a strong foundation for future enhancements such as real-time forecasting, advanced deep learning models, and integration with inventory management systems. Overall, this project contributes toward smarter, data-driven retail operations and supports effective strategic planning for improved business performance.

CHAPTER-12

FUTURE ENHANCEMENTS

FUTURE ENCHANCEMENTS

The Big Mart Sales Prediction project provides a robust framework for forecasting sales, but several enhancements can be implemented to make the system more powerful, accurate, and practical for real-world business applications. These enhancements focus on improving data quality, model performance, operational efficiency, and decision-making support.

1. **Integration of Additional Data Sources**

   Future versions of the project can incorporate external datasets such as promotional campaigns, festive or holiday periods, competitor pricing, regional economic indicators, and weather patterns. Including these factors will help the model account for demand fluctuations caused by seasonal trends, marketing strategies, or external events, resulting in more precise sales predictions.

2. **Real-Time Sales Forecasting**

   By integrating real-time sales data from all outlets, the system can provide dynamic forecasts that adjust as new data becomes available. This will help in proactive inventory management, reducing overstocking or stockouts, and improving responsiveness to changing customer demand.

3. **Advanced Machine Learning Techniques**

   While the current project uses XGBoost and Random Forest, future enhancements can include deep learning models such as Long Short-Term Memory (LSTM) networks or recurrent neural networks (RNNs) for time-series forecasting. Hybrid models combining ensemble methods with neural networks can also be explored to improve accuracy and capture complex sales patterns.

4. **Automated Inventory and Supply Chain Optimization**

   By linking predicted sales with inventory management systems, the project can evolve into a complete operational tool. Future enhancements can include automated stock replenishment suggestions, warehouse optimization, and supply chain planning, ensuring products are available in the right quantity at the right time.

5. **Personalized Promotions and Recommendation Systems**

   Leveraging customer purchase patterns, the system can be enhanced to recommend personalized offers, bundle products, or highlight high-margin items for promotion. This can help increase sales, enhance customer engagement, and optimize marketing strategies.

6. **Dashboard and Visualization Tools**

   Developing an interactive dashboard for managers will allow visualization of sales predictions, top-selling items, and outlet performance in real time. Future enhancements can include alerts for low-stock items, underperforming products, or high-demand periods, making the system a practical decision-support tool.

7. **Scalability for Multi-Region Operations**

   The model can be enhanced to predict sales for multiple regions or countries by including demographic and geographical features. This will enable Big Mart to plan region-specific strategies, optimize distribution, and expand operations efficiently.

8. **Predictive Analytics for Strategic Decision-Making**

   Future enhancements can extend the system to provide insights for strategic decisions, such as opening new outlets, pricing adjustments, or inventory planning. Predictive analytics can help management anticipate market trends and take proactive business actions.

9. **Cloud-Based Deployment and Automation**

   Hosting the system on the cloud and automating data updates and model retraining can make it accessible across multiple locations. This enhancement will allow seamless integration with existing business systems and provide continuous, reliable sales forecasts.

The Big Mart Sales Prediction project provides a strong foundation for forecasting sales across outlets, but there are several opportunities to expand and enhance the system in the future. One of the most significant directions is the **incorporation of external and dynamic data sources**. Currently, the model relies primarily on historical sales and outlet information. By including factors such as seasonal variations, festival periods, promotional campaigns, competitor pricing, and regional economic conditions, the prediction accuracy can be significantly improved. This would allow the model to capture fluctuations in customer behavior that occur due to external influences, making it more robust and reliable.

Another important area of expansion is **real-time sales prediction and dynamic inventory management**. Integrating live sales data streams from all outlets can allow the model to continuously update predictions, enabling immediate adjustments in stock levels and replenishment schedules. This can prevent overstocking or stockouts, reduce operational costs, and ensure that high-demand products are always available. Real-time forecasting can also assist in dynamic pricing strategies, where prices are adjusted based on demand trends, inventory levels, and competitor activity.

The predictive model can further be enhanced using **advanced machine learning and deep learning techniques**. While the current project uses XGBoost and Random Forest, future implementations could employ recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) networks to model time-series trends and seasonal effects in sales data. Hybrid models combining ensemble methods with neural networks could also be explored to improve accuracy for large, complex datasets with non-linear relationships. Additionally, reinforcement learning techniques could be utilized to optimize store-level decision-making, such as stock allocation and promotional strategies.

Personalization and recommendation systems present another area for growth. By analyzing customer purchase patterns at the outlet level, Big Mart could implement targeted promotions and personalized recommendations, boosting sales and customer satisfaction. For example, items that are frequently bought together can be bundled or promoted strategically, and high-margin products can be highlighted during peak shopping periods.

The project can also be extended into a **decision-support dashboard** for managers and executives. Such a system would provide visualizations of predicted sales, top-performing products, and outlet-specific performance in real time. Alerts can be integrated for underperforming items, low-stock situations, and high-demand periods, enabling data-driven operational decisions. Moreover, predictive insights can guide strategic decisions, such as opening new outlets in high-demand locations or planning supply chain logistics more effectively.

Finally, there is potential to **scale the model to multi-country operations**. By including geographical and demographic features, the model could predict sales for outlets in different regions, adapting to local trends, preferences, and purchasing behavior. This would make the system valuable not only for Big Mart but also for other large retail chains looking for a scalable, data-driven approach to sales forecasting.

In conclusion, the future scope of this project is broad, encompassing **integration of external data, real-time forecasting, advanced AI techniques, personalized recommendations, dashboard-based visualization, and global scalability**. These enhancements can transform the sales prediction model from a static forecasting tool into a comprehensive decision-support system, enabling Big Mart to optimize operations, increase profitability, and better serve its customers.

CHAPTER-13

REFERENCES

REFERENCES

[1] Ching Wu Chu and Guoqiang Peter Zhang, “A comparative study of linear and nonlinear models for aggregate retails sales forecasting”, Int. Journal Production Economics, vol. 86, pp. 217- 231, 2003.

[2] Wang, Haoxiang. "Sustainable development and management in consumer electronics using soft computation." Journal of Soft Computing Paradigm (JSCP) 1, no. 01 (2019): 56.- 2.

[3] Suma, V., and Shavige Malleshwara Hills. "Data Mining based Prediction of Demand in Indian Market for Refurbished Electronics." Journal of Soft Computing Paradigm (JSCP) 2, no. 02 (2020): 101- 110

[4] Giuseppe Nunnari, Valeria Nunnari, “Forecasting Monthly Sales Retail Time Series: A Case Study”, Proc. of IEEE Conf. on Business Informatics (CBI), July 2017.

[5]https://halobi.com/blog/sales-forecasting-five-uses/.

[6] Zone-Ching Lin, Wen-Jang Wu, “Multiple Linear Regression Analysis of the Overlay Accuracy Model Zone”, IEEE Trans. On Semiconductor Manufacturing, vol. 12, no. 2, pp. 229 – 237, May 1999.

[7] O. Ajao Isaac, A. Abdullahi Adedeji, I. Raji Ismail, “Polynomial Regression Model of Making Cost Prediction In Mixed Cost Analysis”, Int. Journal on Mathematical Theory and Modeling, vol. 2, no. 2, pp. 14 – 23, 2012.

[8] C. Saunders, A. Gammerman and V. Vovk, “Ridge Regression Learning Algorithm in Dual Variables”, Proc. of Int. Conf. on Machine Learning, pp. 515 – 521, July 1998.IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. 56, NO. 7, JULY 2010 3561.

[9] ”Robust Regression and Lasso”. Huan Xu, Constantine Caramanis, Member, IEEE, and Shie Mannor, Senior Member, IEEE. 2015 International Conference on Industrial Informatics-Computing Technology, Intelligent Technology, Industrial Information Integration.”An improved Adaboost algorithm based on uncertain functions”.Shu Xinqing School of Automation Wuhan University of Technology.Wuhan, China Wang Pan School of the Automation Wuhan University of Technology Wuhan, China.

[10] Xinqing Shu, Pan Wang, “An Improved Adaboost Algorithm based on Uncertain Functions”, Proc. of Int. Conf. on Industrial Informatics – Computing Technology, Intelligent Technology, Industrial Information Integration, Dec. 2015.

[11] A. S. Weigend and N. A. Gershenfeld, “Time series prediction: Forecasting the future and understanding the past”, Addison-Wesley, 1994.

[12] N. S. Arunraj, D. Ahrens, A hybrid seasonal autoregressive integrated moving average and quantile regression for daily food sales forecasting, Int. J. Production Economics 170 (2015) 321-335P

[13] D. Fantazzini, Z. Toktamysova, Forecasting German car sales using Google data and multivariate models, Int. J. Production Economics 170 (2015) 97-135.

[14] X. Yua, Z. Qi, Y. Zhao, Support Vector Regression for Newspaper/Magazine Sales Forecasting, Procedia Computer Science 17 (2013) 1055–1062.

[15] E. Hadavandi, H. Shavandi, A. Ghanbari, An improved sales forecasting approach by the integration of genetic fuzzy systems and data clustering: a Case study of the printed circuit board, Expert Systems with Applications 38 (2011) 9392–9399.

[16] P. A. Castillo, A. Mora, H. Faris, J.J. Merelo, P. GarciaSanchez, A.J. Fernandez-Ares, P. De las Cuevas, M.I. Garcia-Arenas, Applying computational intelligence methods for predicting the sales of newly published books in a real editorial business management environment, Knowledge-Based Systems 115 (2017) 133-151.

[17] R. Majhi, G. Panda and G. Sahoo, “Development and performance evaluation of FLANN based model for forecasting of stock markets”. Expert Systems with Applications, vol. 36, issue 3, part 2, pp. 6800-6808, April 2009.

[18] Pei Chann Chang and Yen-Wen Wang, “Fuzzy Delphi and back propagation model for sales forecasting in PCB industry”, Expert systems with applications, vol. 30,pp. 715-726, 2006.

[19] R. J. Kuo, Tung Lai HU and Zhen Yao Chen “application of radial basis function neural networks for sales forecasting”, Proc. Of Int. Asian Conference on Informatics in control, automation, and robotics, pp. 325- 328, 2009.

[20] R. Majhi, G. Panda, G. Sahoo, and A. Panda, “On the development of Improved Adaptive Models for Efficient Prediction of Stock Indices using Clonal-PSO (CPSO) and PSO Techniques”, International Journal of Business Forecasting and Market Intelligence, vol. 1, no. 1, pp.50-67, 2008.

[21]Suresh K and Praveen O, "Extracting of Patterns Using Mining Methods Over Damped Window," 2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2020, pp. 235-241, DOI: 10.1109/ICIRCA48905.2020.9182893.

[22] Shobha Rani, N., Kavyashree, S., & Harshitha, R. (2020). Object Detection in Natural Scene Images Using Thresholding Techniques. Proceedings of the International Conference on Intelligent Computing and Control Systems, ICICCS 2020, Iciccs, 509–515.

[23] https://www.kaggle.com/brijbhushannanda1979/bigmartsalesdata.